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on

"Biometric Blood Group Classifier using Fingerprinting"

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF ENGINEERING

In

INFORMATION SCIENCE & ENGINEERING

By

Lalana	**1AP22IS020**
Mahammed Kaif H	**1AP22IS023**
Maruthi Reddy B M	**1AP22IS025**
Prerana Hebsur	**1AP22IS036**

Under the guidance of:

Dr. Mithun B N
Prof and Head,
Dept. of ISE

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DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

A P S COLLEGE OF ENGINEERING

Anantha Gnana Gangothri,
NH-209, Kanakapura Road, Somanahalli, Bengaluru-560116.

A.P.S. COLLEGE OF ENGINEERING

(Affiliated to Visvesvaraya Technological University)
Anantha Gnana Gangotri,
NH-209, Kanakapura Road, Bengaluru-560116.

DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

CERTIFICATE

This is Certified that the Project work entitled

"Biometric Blood Group Classifier using Fingerprinting"

is a bonafide work carried out by

Lalana	**1AP22IS020**
Mahammed Kaif H	**1AP22IS023**
Maruthi Reddy B M	**1AP22IS025**
Prerana Hebsur	**1AP22IS036**

In partial fulfilment for the award of degree of Bachelor of Engineering in Information Science & Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2025-2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Project report has been approved as it satisfies the academic requirements concerning the project work prescribed for the Bachelor of Engineering Degree.

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Signature of the Guide	Signature of the Coordinator	Signature of Principal
Dr. Mithun B N	**Dr. Mithun B N**	**Dr K S Anand**
Prof and Head,	Prof and Head,	Principal i/c
Dept. of ISE	Dept. of ISE	

Name of the External **Signature and Date**

1.

2.

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Lalana	**1AP22IS020**
Mahammed Kaif H	**1AP22IS023**
Maruthi Reddy B M	**1AP22IS025**
Prerana Hebsur	**1AP22IS036**

ABSTRACT

Access to accurate medical information is crucial in emergency healthcare; however, individuals who are unconscious, unidentified, or without medical records often face delays that hinder timely treatment. The absence of immediate blood type information can obstruct critical procedures such as transfusions or surgeries. Conventional blood typing relies on laboratory tests or physical documentation, both of which require time and resources creating a significant gap in emergency readiness and patient safety.

The Biometric Blood Group Classifier (BBGC) addresses this challenge by integrating fingerprint-based biometric recognition with blood type retrieval using image similarity techniques. The system uses ORB (Oriented FAST and Rotated BRIEF) feature extraction to analyse distinctive fingerprint ridge patterns and compare them with an encrypted database of pre-labelled fingerprints. Instead of performing biological inference, the system retrieves the blood group linked to the closest-matching fingerprint, ensuring rapid access to essential medical data.

To protect sensitive biometric information, all fingerprint images and metadata are secured using Fernet symmetric-key encryption, guaranteeing confidentiality and integrity. The system's lightweight design, fast matching process, and offline operability make it suitable for varied environments. Accuracy is further improved through image preprocessing, blur detection, and controlled capture conditions, while a mobile-friendly interface enables seamless fingerprint acquisition via standard smartphone cameras.

The BBGC delivers real-time blood group identification and compatible donor information, making it especially beneficial in emergencies. Its applications extend across hospitals, ambulances, rural clinics, disaster response teams, blood donation camps, and forensic units, enabling faster decision-making and improved patient outcomes. By enabling instant retrieval of blood type information through fingerprint-based identification, the BBGC enhances treatment speed, strengthens emergency preparedness, and expands medical accessibility. This project highlights the transformative potential of biometric technologies in building more responsive, secure, and inclusive healthcare systems.

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CHAPTER-1

INTRODUCTION

Biometric systems play a vital role in modern identity verification and security frameworks, offering authentication methods based on unique physiological or behavioural traits. Among the wide range of biometric identifiers, fingerprints remain the most widely adopted due to their uniqueness, stability, ease of acquisition, and universal applicability. Fingerprint recognition has long been utilized in fields such as law enforcement, forensics, digital attendance, mobile device authentication, and banking services. Its reliability and low-cost implementation make it one of the most practical biometric technologies available today. Building on these strengths, the Biometric Blood Group Classifier (BBGC) extends the traditional use of fingerprint recognition by linking biometric identification with critical medical information, demonstrating how fingerprint-based systems can support life-saving applications beyond conventional security and authentication domains.

In the medical domain, blood group information is a critical component of patient identification and emergency readiness. It directly influences life-saving procedures such as trauma management, organ transplantation, and blood transfusion decisions. However, in real-world scenarios, patients often do not carry medical records, and healthcare providers may not have immediate access to blood group data especially in the case of unconscious individuals, accident victims, or unidentified persons. The delay involved in laboratory testing can significantly impact medical response time, potentially putting lives at risk.

The project, aims to bridge this accessibility gap by linking biometric identity with essential medical information. Instead of predicting blood groups through biological analysis- an approach that is scientifically complex and inconsistent - this system retrieves a person’s blood group through fingerprint similarity matching. Each stored fingerprint image in the system is labelled with a known blood group. When a new fingerprint is captured or uploaded, the system compares its unique ridge patterns using ORB (Oriented FAST and Rotated BRIEF) feature extraction combined with Hamming-distance matching to identify the closest match in the encrypted fingerprint database.

To ensure data privacy and security, the system incorporates Fernet symmetric-key encryption, protecting all stored fingerprint images and metadata from unauthorized access. A built-in blur detection mechanism evaluates the quality of uploaded images to prevent inaccurate matching

caused by unclear fingerprints. The entire workflow is implemented as a Flask-based web application, providing both an administrative dashboard for managing labelled fingerprints and a mobile-friendly user interface for real-time fingerprint capture and prediction.

The project demonstrates how biometric technologies, when combined with secure data handling and efficient image processing, can support critical medical operations. Its offline capability, lightweight implementation, and rapid prediction time make it suitable for use in emergency medical units, rural healthcare centres, ambulances, forensic environments, military operations, and blood donation drives. Beyond emergency support, the system lays the foundation for future developments such as integrating machine learning models, expanding biometric modalities, and connecting with national healthcare databases.

1.1 Statement of the Problem

During medical emergencies, especially in accident scenarios or unconscious-patient cases, healthcare workers often lack access to essential medical history, particularly blood group information. Although biometric systems exist for identity verification, none of them directly link fingerprint data with blood group information. Medical databases are usually centralized, require authentication, and may not be accessible in remote locations or in real time.

Challenges identified include:

- Lack of instant availability of patient blood group records in emergencies.
- Dependence on laboratory testing, which takes time and may delay treatment.
- Absence of systems that link fingerprint biometrics with blood group information.
- Security concerns when storing biometric data without proper encryption.
- Requirement for offline functioning in rural or field environments with no internet.

Thus, there is a need for a secure, fast, offline-capable system that retrieves a person's blood group by matching their fingerprint against a locally stored encrypted fingerprint repository using similarity-based classification.

1.2 Objectives of the Problem

The main objective of this project is to develop a secure and intelligent biometric system that can identify a person's blood group using fingerprint image similarity and ridge-pattern matching. The system uses ORB-based key point extraction and descriptor comparison to match newly scanned fingerprints with an encrypted database, ensuring fast, accurate, and offline-capable blood group prediction. All fingerprint images and metadata are protected using Fernet symmetric encryption to

maintain privacy and prevent unauthorized access. Another objective is to create a highly usable and efficient platform that works seamlessly for both administrators and end users. The system validates fingerprint quality through blur detection, enables quick labeling and prediction through an intuitive interface, and delivers results within 1–2 seconds even without internet connectivity. The design supports smooth scalability, allowing more fingerprint samples to be added over time while maintaining performance and accuracy.

A further objective is to support healthcare operations by displaying compatible blood donor groups and providing fast access to critical information needed during emergencies. The system is designed for practical deployment in ambulances, rural clinics, blood donation camps, and emergency response units. In the future, the project aims to integrate advanced machine learning classifiers, expand the encrypted fingerprint database, and enhance its role as a non-invasive biometric tool in medical assistance and patient identification.

1.3 Scope of the Problem

The scope of this project is extensive, covering both the operational functionality and the real-world applicability of the BBGC. At its core, the system is designed as a retrieval-based biometric solution rather than a biological prediction model, meaning it identifies a user's blood group by matching their fingerprint against a pre-labelled, encrypted fingerprint database. This approach ensures accuracy, practical feasibility, and ethical compliance, as it avoids speculative biological inference. The system supports two primary modes of operation: administrative functions, where authorized personnel can upload and label fingerprints with corresponding blood groups, and user-facing functions that allow individuals to capture and upload new fingerprint images for blood group prediction. Functionally, the system encompasses fingerprint acquisition through mobile cameras or webcams, preprocessing for noise removal and blur validation, ORB key-point extraction and descriptor generation, secure encryption of biometric templates and metadata, and robust feature matching using Hamming distance and ratio tests. The output includes not only the predicted blood group but also key performance indicators such as good-match count, total matches, and blur variance to validate confidence levels.

From an application perspective, the system is designed to serve a wide range of critical environments, including hospitals, emergency medical services, blood donation camps, disaster management units for unidentified victims, military field operations requiring offline access, and forensic departments that require rapid retrieval of identity-linked medical data. Its lightweight

architecture, offline capability, and minimal hardware requirements make it suitable for deployment in resource-constrained regions. Practically, the BBGC system helps eliminate delays associated with conventional blood typing tests, accelerates emergency decision-making, and greatly enhances preparedness in life-threatening situations where every second matters. Additionally, the system enables improved accessibility to essential medical information in remote or rural areas, thereby supporting broader digital health initiatives. Strong encryption mechanisms ensure that all sensitive biometric and medical data are fully protected, maintaining user privacy and adhering to security standards.

Overall, the scope of this project establishes a comprehensive, secure, and efficient biometric identification framework that seamlessly integrates fingerprint recognition with essential medical information retrieval. It defines the entire operational workflow from image capture, preprocessing, and ORB-based feature extraction to encrypted template management, feature matching, and donor-compatibility display while supporting both backend administrative control and real-time user interaction. With its broad applicability, strong security features, and high practical impact, the BBGC significantly improves diagnostic efficiency, reduces treatment delays, and contributes to more responsive, reliable, and inclusive healthcare delivery systems.

1.4 Methodology

The project adopts a comprehensive and structured methodology that seamlessly integrates biometric image processing, cryptographic security, ORB-based feature extraction, and web-driven deployment to deliver a reliable fingerprint-based blood group retrieval system. The development process begins with Requirement Analysis and System Planning, where fingerprint ridge structures and their biometric properties are studied to understand how they can be used for feature extraction and matching. During this stage, the architectural blueprint of the system is defined, including the separation of admin and user workflows, encrypted storage mechanisms, and the design of a responsive user interface. ORB is selected as the core feature extraction technique due to its computational efficiency, robustness to rotation, and suitability for lightweight, real-time applications. Critical thresholds such as blur variance and minimum match scores are established to ensure prediction accuracy and avoid weak or unreliable matches.

In Image Acquisition and Preprocessing, fingerprints are captured through smartphone cameras or uploaded via the web interface, ensuring flexibility across devices. Each image undergoes standardized preprocessing where it is converted to grayscale for uniform analysis and evaluated

for blur variance using Laplacian-based metrics. This early rejection of low-quality images is essential to prevent processing overhead and to maintain high reliability during the matching phase. The subsequent Feature Extraction phase employs ORB's FAST key-point detector and BRIEF descriptor logic to identify ridge endings, bifurcations, and distinctive fingerprint structures. Up to 1,500 key points are extracted to create a strong, distinctive descriptor set that captures essential fingerprint characteristics, ensuring robustness even when minor distortions or rotations occur during capture.

The Encrypted Storage phase is fundamental to the security architecture of the system. All fingerprint images are encrypted using Fernet AES-based cryptographic tokens before being saved, ensuring that no raw biometric data is ever stored in the system. Metadata including file names, blood group labels, timestamps, and blur scores is also encoded and stored in an encrypted CSV file. This strict separation of encrypted images and encrypted metadata, combined with controlled file-system access, ensures full compliance with biometric data protection standards and prevents unauthorized access or tampering.

During Index Building and Matching, the system constructs an in-memory ORB descriptor index each time a new labelled fingerprint is added to the database. This index is dynamically rebuilt to incorporate new entries and ensure that the matching process always uses the most updated dataset. Descriptor matching is performed using a Brute-Force Matcher based on Hamming distance, which compares each bit-level descriptor efficiently. Lowe's Ratio Test is applied to filter ambiguous or weak matches, retaining only the strongest correspondences. The fingerprint with the highest number of good matches is considered the closest match, and its match score is compared against the pre-defined confidence threshold (typically 12) to determine prediction validity.

In the Prediction and Output phase, when a fingerprint meets the required confidence threshold, the system returns essential information such as predicted blood group, number of good matches, total matches evaluated, matched file reference, blur score, and a compatible donor list based on standard ABO/Rh medical guidelines. If the match confidence falls below acceptable limits, the system communicates clear and actionable error messages to guide the user. This ensures that only reliable predictions are delivered, maintaining system integrity and preventing misclassification. The methodology concludes with User Interface Development, which includes a web-based admin dashboard and a mobile-friendly interface for real-time use. The admin dashboard allows authorized personnel to upload and label fingerprints, manage the encrypted template database, and manually

rebuild the descriptor index when required. The mobile UI, designed for emergency and field operations, enables quick fingerprint capture using smartphone cameras, followed by instant processing and prediction display. Both interfaces are built to be intuitive, responsive, and secure, ensuring smooth operation across multiple platforms and environments. Together, these seven phases form a complete end-to-end workflow that ensures accurate biometric matching, secure data handling, scalable system functionality, and practical usability across healthcare, forensic, and emergency response settings.

1.5 Related Work

Research in biometric science has extensively explored the relationship between fingerprint ridge patterns and physiological traits such as blood group, gender, and ethnicity, with numerous studies attempting to correlate ridge features such as loops, whorls, and arches with ABO/Rh blood types. Although some statistical associations have been observed, the findings vary widely due to differences in population samples, fingerprint acquisition conditions, and analytical methodologies, making these correlations insufficient for accurate biological blood-group prediction. Prior works by researchers such as Prateek Rastogi, Yeswanth K., and Vijaykumar Patil focused on large-scale statistical analysis of fingerprint ridge flows across diverse demographics, contributing useful datasets but failing to account for distortions, image quality issues, and environmental variability. Importantly, these studies do not propose automated or real-time classification systems and offer associations rather than clinically reliable diagnostic rules.

Machine learning approaches including CNN- and SVM-based models have demonstrated success in classifying fingerprint patterns and identifying individuals, but they rely heavily on large, labelled datasets, substantial computational resources, and lack essential components such as secure storage, encryption, and medical workflow integration. These models do not link fingerprint characteristics to blood group information and are therefore unsuitable for real-time medical deployment. Traditional biometric matching systems utilizing ORB, SIFT, SURF, or Minutiae-based algorithms excel in identity verification by analysing geometric ridge patterns and remain widely adopted due to their robustness to finger orientation and efficiency in descriptor matching. However, these systems focus solely on personal authentication and do not associate biometric templates with medical datasets, nor do they incorporate end-to-end security mechanisms required for handling sensitive medical and biometric data. This project addresses a significant and previously unexplored gap by introducing a fully functional, privacy-preserving, and real-time fingerprint-based blood group retrieval system that bridges biometrics and healthcare information

systems. Unlike earlier studies that focus on correlation or traditional matching tools that emphasize identity verification alone, this system implements a retrieval-based mechanism that maps fingerprints to pre-labelled blood group records using ORB feature matching rather than attempting biological prediction. The system ensures complete privacy by encrypting both fingerprint templates and metadata through AES-based mechanisms, making it secure for clinical use. Its lightweight Flask-based architecture enables deployment across smartphones, computers, and resource-limited medical environments. Additional advancements include the integration of blur-detection algorithms to prevent degraded images from compromising prediction accuracy and support for both administrative enrolment and user-side prediction through an intuitive, mobile-friendly interface. Collectively, these contributions position the system as a novel and practical solution capable of delivering rapid, secure, and medically relevant fingerprint-based information retrieval in emergency healthcare settings.

1.6 Organization of Report

This project report is structured into eight chapters, followed by references and appendices. Each chapter presents the concepts, design, implementation, and evaluation of the Biometric Blood Group Classifier Using Fingerprint Images.

- Chapter 1 – Introduction
  Provides an overview of the project, including motivation, problem statement, objectives, scope, methodology, related work, and a summary of the report structure.
- Chapter 2 – Literature Survey
  Reviews existing research on biometric fingerprint analysis, blood group correlation studies, image-processing techniques, ORB-based matching, and secure biometric systems. It highlights gaps in current work and shows how this project offers a secure, retrieval-based, non-invasive solution.
- Chapter 3 – Software Requirements Specification (SRS)
  Defines the system's requirements including hardware/software needs, user characteristics, functional and non-functional requirements, system constraints, and assumptions.
- Chapter 4 – Design
  Describes the system design through architectural diagrams, block diagrams, flow diagrams, and UML models such as class, sequence, and activity diagrams. It explains module interactions and system data flow.

- Chapter 5 – Implementation

  Explains the development process, environment setup, directory structure, preprocessing workflow, ORB extraction, encrypted storage, matching logic, Flask backend, and UI features for both admin and mobile users.
- Chapter 6 – Testing

  Presents the testing strategy, including functional, performance, security, usability, compatibility, and edge-case testing. Includes test plans, scenarios, expected results, and actual outcomes.
- Chapter 7 – Snapshots

  Displays screenshots of the working system admin dashboard, mobile capture interface, encrypted storage views, blur detection, ORB matching results, and blood group prediction output.
- Chapter 8 – Conclusion and Future Scope

  Summarizes project outcomes, discusses limitations, and suggests improvements such as integrating deep learning, expanding the dataset, enhancing security, and enabling cloud or offline deployment.
- References

  Lists all textbooks, journals, websites, and research papers used in the project.
- Appendix A, B, C, D

  Provide extended workflows, additional diagrams, sample outputs, ORB visualizations, dataset descriptions, logs, and supplementary material that supports the main chapters, Abbreviations and full forms, Authorization Certificate from Medical Official.

CHAPTER-2

LITERATURE SURVEY

The literature survey presents a comprehensive review of existing research related to biometric identification, fingerprint analysis, and blood group association techniques. It examines prior studies, methodologies, and technological advancements to identify limitations in current approaches and establish the research gap addressed by the proposed system.

As biometric technologies advance, their impact on healthcare, security, and digital identity systems has become increasingly significant. Fingerprint recognition, one of the most reliable and widely adopted biometric modalities, continues to play a vital role in authentication due to its uniqueness, permanence, and ease of acquisition. Alongside these developments, researchers have explored the relationship between dermatoglyphic patterns and biological attributes such as blood group, disease susceptibility, and genetic traits. Early fingerprint anthropology studies reported observable tendencies in ridge patterns across different ABO blood groups, indicating that fingerprints may encode subtle biological information useful for non-invasive medical applications. Although these findings were not sufficient for deterministic blood group prediction, they laid the foundation for integrating biometric imaging with medical data retrieval systems [1].

With the rapid progress of image-processing techniques and machine-learning methodologies, researchers began exploring automated systems for blood-group identification. A significant body of work focused on image-based interpretation of blood-sample reactions using spectrophotometry, optical sensing, and reaction-well imaging. These systems demonstrated high accuracy in detecting agglutination patterns and significantly reduced reliance on manual laboratory observation [2,3]. However, these approaches remained inherently invasive, requiring physical blood samples and controlled environments. This limitation created a strong need for alternative, non-invasive mechanisms capable of retrieving blood-group information without laboratory equipment especially in emergency scenarios where conventional tests may be delayed.

In parallel, fingerprint recognition research expanded significantly, incorporating robust algorithms such as DWT, Gabor filtering, HOG descriptors, minutiae extraction, SIFT, and ORB. These methods enhanced matching accuracy under varying lighting conditions, sensor types, and image resolutions. Several works also addressed the increasing prevalence of mobile biometric acquisition, proposing descriptors capable of handling noise, blur, rotation, and compression artifacts typically associated with smartphone-captured images [4].

More recent efforts shifted towards directly associating fingerprint features with blood group prediction. Studies employing wavelet transforms, ridge-density analysis, and handcrafted feature extraction reported moderate success, demonstrating that fingerprints contain patterns that can be correlated with ABO blood types [5]. Although deep-learning-based fingerprint classifiers achieved higher accuracy, they were constrained by the scarcity of large labelled datasets and the computational demands of training and inference. Moreover, existing models generally lacked strong security mechanisms, leaving biometric data vulnerable to exposure or misuse. This gap highlighted the need for systems that not only perform accurate fingerprint matching but also prioritize encryption, template protection, and secure metadata handling.

Security and privacy concerns have therefore emerged as critical considerations in biometric research. Traditional systems often store raw fingerprints or unprotected templates, posing risks of identity leakage and unauthorized access. Several studies stress the necessity of encrypted storage, privacy-preserving architectures, and controlled access mechanisms to safeguard sensitive biometric information in both medical and non-medical applications [6]. The absence of unified frameworks combining secure storage, non-invasive prediction, and real-time usability underscored the need for solutions capable of integrating biometric recognition with medical data retrieval while maintaining high levels of security and reliability.

As research into biometric prediction systems expanded, several studies explored hybrid and feature-fusion techniques to improve the stability and accuracy of fingerprint-based classification. Works involving Discrete Wavelet Transform (DWT), Gabor filters, and Local Binary Patterns (LBP) demonstrated that combining global ridge flow with local texture patterns improved robustness against variations in lighting and pressure during fingerprint capture [7]. These combined approaches were particularly useful for low-quality images captured on mobile devices, reflecting the growing demand for portable and field-ready biometric systems. Some researchers further enhanced these models through statistical filtering and noise-reduction techniques, emphasizing the importance of image preprocessing in improving the reliability of downstream feature extraction.

Later studies incorporated machine-learning models such as Support Vector Machines (SVMs), K-Nearest Neighbour (KNN), and Random Forest classifiers to identify fingerprint classes and, in some cases, associate these classes with blood-group information. Although these models showed measurable improvements over traditional handcrafted approaches, their performance was strongly dependent on dataset size and variability. Limited availability of labelled fingerprint datasets especially datasets linking ridge patterns to blood groups posed a significant challenge for achieving

generalizable results [8]. This dataset scarcity also made it difficult to evaluate models across different demographics, highlighting the need for systems that rely not on prediction, but on *retrieval-based approaches*, where fingerprints are matched against labelled templates rather than used for direct classification.

Deep-learning-based fingerprint recognition marked another major milestone in the literature. Convolutional neural networks (CNNs), residual networks, and Siamese architectures were applied to fingerprint matching and class prediction, offering substantial improvements in handling distortion, rotation, and sensor inconsistencies. These models excelled at extracting deep hierarchical features from fingerprint images, effectively capturing ridge continuity and minutiae-like structures. However, researchers consistently noted the computational cost associated with training and deploying these models, particularly in real-time or resource-constrained environments. Their dependence on GPU acceleration and large, diverse datasets limited their feasibility in emergency medical settings where rapid, lightweight deployment is essential [9].

A significant branch of research also focused on securing biometric systems through encryption, template transformation, and protected key generation. Several papers emphasized the vulnerabilities inherent in storing raw biometric templates, noting that fingerprints cannot be “changed” if compromised making secure storage a critical requirement. Techniques involving AES encryption, cancellable biometrics, and secure metadata storage were highlighted as essential components of modern biometric frameworks. However, while these studies explored encryption independently, they did not integrate secure storage with real-time fingerprint matching or medical data retrieval, leaving a gap for fully protected end-to-end biometric systems [10].

In addition to biometric recognition and encryption, several works examined the operational challenges of using image-based systems in real-world settings. Studies on blur detection, illumination correction, and noise filtering highlighted the importance of validating input image quality before processing. Poor-quality fingerprints lead to unreliable feature extraction and false matching results, potentially compromising system accuracy. Researchers proposed various quality-assessment techniques including Laplacian variance, wavelet-based sharpness analysis, and local contrast evaluation to systematically reject low-quality fingerprints [11]. These findings reinforced the need for robust pre-processing pipelines, especially for systems relying on mobile-captured fingerprint images. Researchers also emphasized the importance of scalability and feature-indexing in biometric identification systems. As fingerprint databases grow, efficient descriptor indexing becomes crucial for maintaining low prediction latency. Prior works introduced indexing structures

such as KD-trees, LSH (Locality-Sensitive Hashing), and clustering-based feature organization to optimize nearest-neighbour searches in large databases. These studies highlighted that retrieval-based systems must continuously update feature indices as new fingerprints are added, ensuring that matching remains fast and accurate across expanding datasets [12]. This insight directly informs the indexing strategy adopted in the present project.

Beyond identity recognition, some studies investigated the integration of biometrics with healthcare workflows. These works emphasized the value of combining secure biometric identification with patient information retrieval in hospitals, emergency services, rural clinics, and forensic environments. Systems capable of linking biometric inputs such as fingerprints or facial images—to essential medical data were shown to significantly improve response times in critical situations, reduce manual errors, and enhance accessibility to patient information even in the absence of physical records [13]. However, existing systems were often too complex, expensive, or lacking in privacy safeguards to be deployed in smaller healthcare facilities.

More recent literature proposed cloud-based and mobile biometric systems that leverage lightweight algorithms for real-time deployment. These studies illustrated the feasibility of running fingerprint recognition pipelines on smartphones, enabling decentralized biometric authentication without requiring large servers or high-end hardware. These advances demonstrated that mobile-ready systems could operate efficiently using compact feature descriptors such as ORB and BRIEF, which offered strong performance despite their lower computational requirements [14]. This shift towards lightweight, deployable biometrics strongly influenced the design decisions of the present system.

Finally, several papers emphasized the practical challenges faced by emergency medical teams when attempting to rapidly identify individuals and determine their blood group. Traditional methods such as serology-based blood typing are accurate but slow, invasive, and dependent on laboratory facilities. In critical situations involving unconscious, unidentified, or unaccompanied patients, delays in acquiring blood-group information can significantly affect treatment outcomes Despite attempts to automate aspects of blood typing, no prior system successfully combined biometric matching with encrypted medical data retrieval, leaving a crucial gap in emergency healthcare technology [15].

CHAPTER-3

REQUIREMENTS SPECIFICATION

The Software Requirements Specification outlines the essential components, constraints, and operational behaviours required to implement the fingerprint-based blood group classification system. It defines the functional and non-functional requirements needed to ensure accurate biometric processing, secure data handling, and efficient system performance. The SRS establishes hardware and software prerequisites, user-level interactions, and system limitations. It also specifies encryption standards, image-processing guidelines, and matching criteria. Overall, these requirements ensure that the application remains reliable, scalable, and suitable for deployment across diverse medical and emergency environments.

3.1 Hardware Requirements

The hardware requirements define the minimum and recommended specifications necessary for operating the system effectively.

- Processor: Dual-Core CPU (1.6 GHz or above).
  Required for basic execution of Flask server and ORB computations.
- RAM: 4 GB
  Ensures smooth image processing and encryption/decryption routines.
- Storage: 2 GB free space
  Used for encrypted fingerprint templates and metadata storage.
- Camera / Fingerprint Source:
  Smartphone camera (8MP or above)
  USB webcam
  Required for capturing fingerprint images through the mobile interface.
- Display:
  Minimum 720p resolution for UI usage.
  Ensures visibility of admin panel functionalities.
  The recommended Hardware Requirements for deployment follows:
- Processor: Quad-Core CPU (2.0 GHz or above)
- RAM: 8 GB
- Storage: SSD for faster read/write operations
- Camera: 12MP+ mobile camera for higher clarity fingerprint captures

3.2 Software Requirements

Operating System

- Windows 10/11
- Linux

Programming Language

- Python 3.10+

Frameworks and Libraries

- Flask Framework – For routing, UI rendering, and API handling.
- OpenCV – For image decoding, ORB feature extraction, blur detection, and matching.
- NumPy – For numerical operations and image array handling.
- Cryptography (Fernet) – For AES-based encryption and secure data storage.
- Jinja2 – Template engine for HTML rendering.
- Werkzeug – For secure file handling.
- UUID, CSV, Datetime Modules – For metadata structuring.

Execution Requirements for the project follows:

- Python virtual environment
- Local or hosted Flask server
- Browser for accessing UI

3.3 Functional Requirements

The functional requirements of the system outline all essential operations needed to ensure accurate, secure, and efficient fingerprint-based blood group classification. The system must first provide a robust image upload mechanism that allows both administrators and general users to submit fingerprint images either through the admin panel on a desktop browser or via the mobile capture interface on a smartphone. To maintain data consistency, the uploaded files must undergo strict validation, including checking that the file format is either JPEG or PNG, confirming that the file is not corrupted, and ensuring that it can be successfully decoded into a grayscale format for further processing. Once a fingerprint is uploaded, the system performs fingerprint quality validation, where it computes the blur variance using the Laplacian operator to assess image sharpness. Any fingerprint falling below the defined blur threshold is automatically rejected to prevent poor-quality images from affecting the accuracy of the matching process, and the system must clearly notify the user that the image is too blurry or unfocused.

Following successful validation, the system proceeds with ORB feature extraction, in which it detects key points using the FAST algorithm and generates binary descriptors using the BRIEF technique. This extraction is mandatory not only for all newly uploaded fingerprints during prediction but also for stored fingerprint templates during index rebuilding whenever new labelled samples are added. To guarantee biometric privacy and comply with secure data-handling practices, the system incorporates encrypted storage, where all fingerprint images and their associated metadata such as filename, blood group, timestamp, and blur score are encrypted using Fernet symmetric AES-based encryption. These encrypted assets are stored in dedicated directories: fingerprints_enc/ for labelled templates and incoming_enc/ for prediction uploads, ensuring that sensitive biometric information remains protected even at the file-system level.

During prediction, the system engages in matching and classification, where the ORB descriptors of the uploaded fingerprint are compared with all stored templates using a Hamming-distance Brute Force Matcher. A prediction is generated only when the number of "good matches" meets or exceeds a predefined threshold, ensuring that classification is both confident and reliable. The system must output comprehensive prediction details, including the predicted blood group, the number of good matches, the total matches evaluated, and the filename corresponding to the best-matching fingerprint. Additionally, the system is required to compute and present a donor compatibility list, outlining all acceptable donor types based on medical ABO and Rh transfusion rules, thereby making the output medically useful and actionable.

To support usability, the system must provide an intuitive user interface comprising two components: an admin panel that enables fingerprint labelling, encrypted database viewing, and ORB index rebuilding, and a mobile interface that facilitates real-time fingerprint capture and instant blood group prediction. Across all platforms, the UI must present results clearly and immediately for smooth user experience. Finally, the system must include comprehensive error-handling capabilities, generating descriptive and meaningful error messages for cases such as blurry or low-quality images, unsupported or corrupted file formats, low-confidence matches, absence of labelled fingerprints in the database, or any encryption/decryption failure. These functional requirements collectively ensure that the system operates reliably, secures biometric data, supports medical decision-making, and remains user-friendly across different environments and usage scenarios.

3.4 Non-Functional Requirements

- Security

  All fingerprint images and metadata must remain encrypted at rest and during processing. Decryption keys must be protected and inaccessible to unauthorized users.

- Performance

  The system should process and return predictions within 1–3 seconds. ORB feature extraction and matching should run efficiently under typical load.

- Scalability

  Support increasing numbers of fingerprint templates without major performance degradation. Allow expansion of the fingerprint dataset and addition of machine learning models.

- Usability

  Provide a clean, intuitive, and responsive interface for both admin and mobile users. Offer clear instructional cues for capturing high-quality fingerprint images.

- Reliability

  Reject low-quality or unreadable fingerprint images automatically. Ensure consistent matching results under varied lighting and capture conditions.

- Portability

  Run seamlessly across different operating systems (Windows, Linux, macOS). Support all major modern browsers for UI access.

CHAPTER-4

DESIGN

The BBGC design ensures accurate fingerprint processing, secure data storage, and dependable blood group retrieval through a modular workflow. It integrates ORB feature extraction, blur-quality validation, encrypted template storage, and descriptor-based matching into a streamlined architecture that maintains both reliability and efficiency. Each component operates independently yet cohesively, enabling fast prediction and robust error handling. The design also supports scalability, allowing new fingerprint templates or advanced models to be added without restructuring the system. This ensures high performance, adaptability, and consistent results across diverse healthcare and emergency scenarios.

4.1 System Architecture

The system architecture is designed to support secure, efficient, and scalable fingerprint-based blood group classification, enabling users to interact seamlessly through both mobile devices and desktop systems. Users access the application via a web or mobile interface built using HTML and Flask, where they upload fingerprint images for processing. This frontend interface performs initial input validation, ensuring that uploaded files meet basic format requirements before forwarding them to the backend. It is intentionally lightweight and responsive, making it suitable for emergency scenarios, remote medical camps, and general healthcare environments where quick access to medical information is essential. At the core of the system operates a Python-based Flask backend, which manages all request routing, decision-making processes, and communication between internal modules. The backend oversees fingerprint validation, blur detection, encryption workflows, ORB feature extraction, similarity matching, donor compatibility generation, and error handling, serving as the central controller that coordinates every stage of fingerprint processing.

Once a fingerprint is received, the preprocessing module performs essential image transformations such as grayscale conversion, noise removal, and blur evaluation to ensure that only high-quality inputs proceed to feature extraction. The ORB feature extraction module then identifies key points and generates binary descriptors using the Oriented FAST and Rotated BRIEF technique, converting the fingerprint into a compact, repeatable, and discriminative numerical signature used for classification. These descriptors are passed to the matching engine, which compares them against stored fingerprint templates using Hamming-distance–based matching along with ratio tests to filter out unreliable matches. The engine then determines the best match, retrieves the corresponding blood group label, and calculates the confidence level. To ensure privacy and data protection, the system

stores all fingerprint images in encrypted .bin format using Fernet AES encryption, while metadata including blood group, timestamp, and blur score is also encrypted and stored within a secure metadata repository. This encrypted storage system functions as the main biometric vault, maintaining strict confidentiality and preventing unauthorized access.

The overall request-handling process is streamlined for efficiency: a user submits a fingerprint, the backend validates and preprocesses the image, ORB descriptors are extracted, template matching is performed, and the resulting prediction is returned to the user. Reliability is reinforced through multiple safeguards, including automatic rejection of blurry or corrupted fingerprints and fallback responses when match confidence is low, ensuring that incorrect or uncertain predictions are never delivered. The architecture is also highly scalable and extensible; new fingerprint templates can be added at any time without interrupting system functionality, and the ORB index rebuilds automatically to incorporate new data. Its modular design allows for future enhancements such as integrating CNN or deep-learning–based fingerprint recognition, expanding databases, and enabling cloud-based deployment or load-balanced setups. Together, these components create a secure, accurate, and flexible biometric classification system capable of supporting real-time medical decision-making across a wide range of environments.

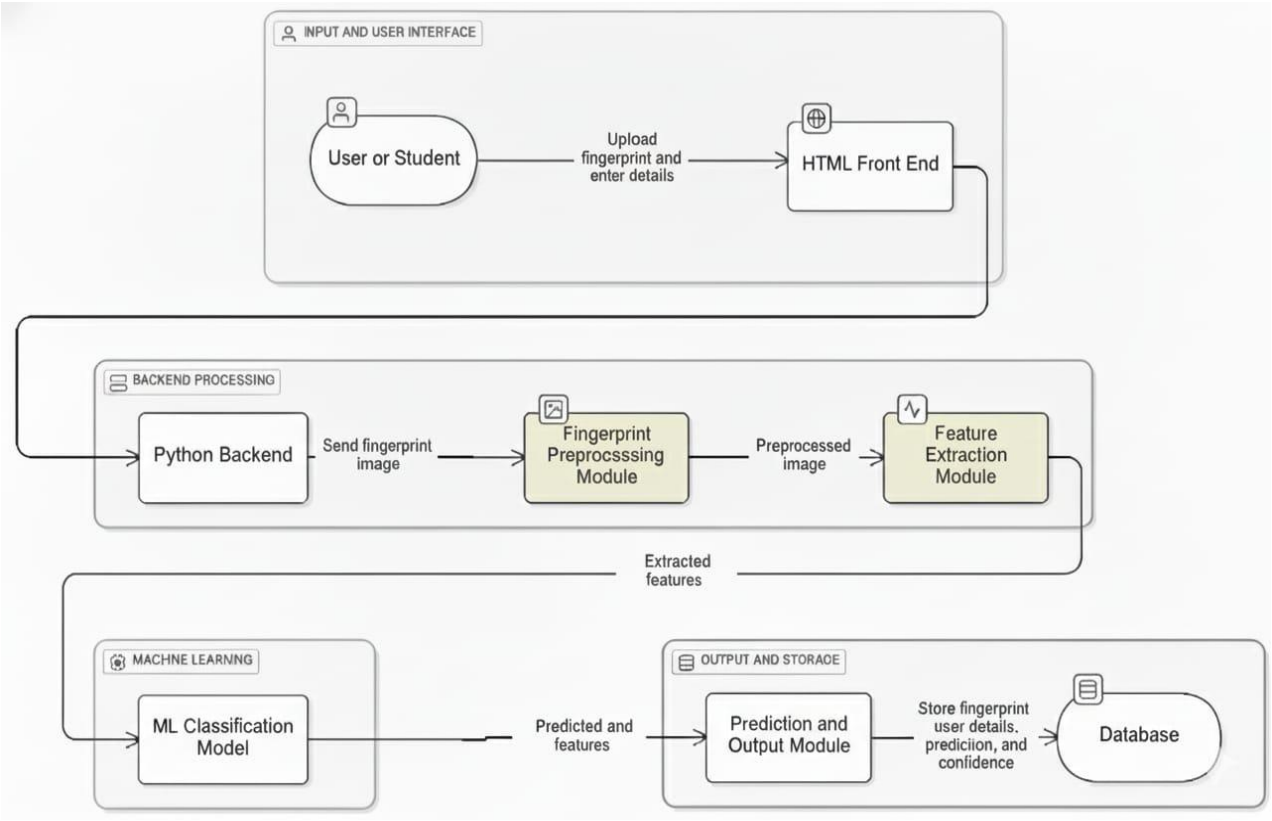


Figure 4.1 BBGC architecture diagram

Figure 4.1 illustrates the overall architecture of the Biometric Blood Group Classifier. Users interact with the system through a web-based or mobile interface to upload fingerprint images, which are routed to the Python backend for processing. The backend handles preprocessing, ORB feature extraction, encryption, and matching operations. The preprocessing module filters out low-quality images, while the feature extraction module generates distinctive ORB descriptors that are used by the matching component to identify the closest fingerprint stored in the encrypted database.

Once the best match is determined, the system retrieves the associated blood group, formats the prediction, and returns it to the user interface. Encrypted storage safeguards all fingerprint templates and metadata, ensuring data privacy and integrity. The modular and scalable design supports high performance across varied conditions and allows for future enhancements such as advanced machine-learning models. Overall, the architecture ensures secure, accurate, and real-time blood group identification suitable for healthcare, forensic, and emergency scenarios.

4.2 Flow diagram

The flow diagram in Figure 4.2 provides a comprehensive representation of the complete operational pipeline for the fingerprint-based blood group identification system, beginning with the establishment of a secure connection between the mobile application and the PC server. This connection achieved through USB, cable, mobile hotspot, or Wi-Fi acts as the foundational communication channel, ensuring that the transfer of biometric data takes place in a controlled and protected environment. Once connectivity is established, the mobile application initiates the fingerprint capture phase, guiding users with visual cues to correctly position their finger for optimal clarity, focus, and lighting. The quality of this captured image is critical, as it forms the primary biometric input for all subsequent processing stages. Immediately after capture, the system enforces stringent data-security measures by performing AES-based encryption on the fingerprint image, effectively locking the data into an unreadable state so that only the authorized server holding the appropriate decryption key may access it. This encryption step ensures full protection of biometric information even before it is transmitted.

Following encryption, the fingerprint image is transferred to the Python backend server through an HTTPS-secured communication channel, where SSL/TLS encryption maintains confidentiality and prevents any unauthorized interception or tampering during transmission. Once received by the server, the encrypted fingerprint undergoes decryption and is then forwarded to the image processing and feature extraction module. Here, the system analyses the fingerprint’s ridge flow, local ridge discontinuities, minutiae points, or ORB-based descriptors, converting it into a numerical signature that uniquely characterizes the fingerprint. These extracted features are then input into the matching

engine, which compares them against all stored fingerprint templates within the encrypted database. Using similarity metrics and descriptor-matching algorithms, the engine determines the closest matching fingerprint entry, thereby identifying the corresponding stored user profile.

After confirming the best match, the system retrieves the blood group label associated with the matched fingerprint, enabling a fast, identity-linked method of obtaining crucial medical data without requiring traditional laboratory testing. This predicted blood group, along with additional details such as the number of good matches, blur score, and compatible donor list, is then processed for secure output. The final results undergo another stage of encryption to preserve confidentiality before being transmitted back to the mobile device over a secure, SSL-protected channel. Upon arrival, the mobile interface decrypts and displays the predicted blood group clearly and immediately, enabling real-time and reliable access to essential medical information. This completes the workflow, showcasing how the system integrates secure data transfer, biometric processing, encryption mechanisms, and pattern-matching logic to deliver fast, accurate, and privacy-preserving medical decision support, especially suited for emergency environments, field operations, and healthcare scenarios requiring rapid response.

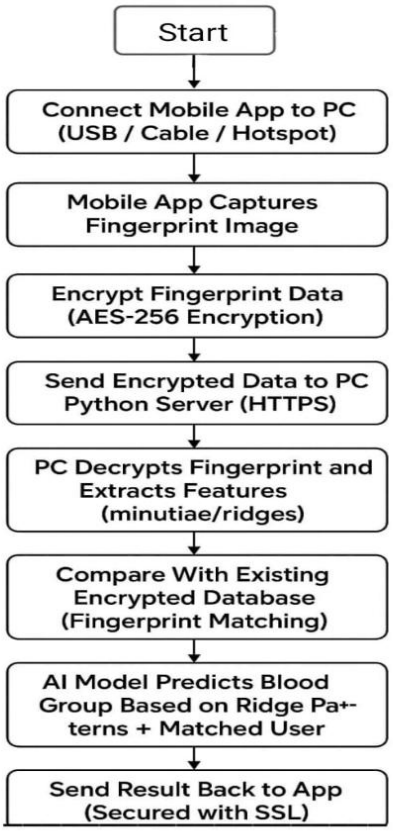


Figure 4.2 Flow diagram for BBGC

4.3 Block Diagram

Figure 4.3 illustrates the block diagram of the Biometric Blood Group Classification system, providing a comprehensive overview of the core components and the sequential flow of data from fingerprint acquisition to final result display. The process begins with the user, who initiates the workflow by placing their finger on a fingerprint scanner or capture device. This user interaction serves as the foundational input for the entire system, as the accuracy and clarity of the captured fingerprint significantly influence downstream processing. The fingerprint scanner then converts the physical ridge and minutiae patterns into a high-resolution digital image, ensuring that the unique biometric traits are preserved for further analysis. Once captured, the image is forwarded to the preprocessing module, where it undergoes a series of enhancement operations including grayscale conversion, noise reduction, normalization, and blur detection. These preprocessing steps refine the fingerprint image quality, enabling reliable feature extraction and preventing poor-quality inputs from degrading the system's prediction accuracy.

After preprocessing, the refined fingerprint enters the blood group classification module, where the system extracts key features such as ridge flow, minutiae distribution, or ORB descriptors and compares them with stored templates or a trained model to determine the corresponding blood group. The classifier evaluates similarities or learned associations and generates a predicted blood group label for the input fingerprint. The system database plays a central role in this workflow by storing fingerprint templates, feature vectors, associated blood group labels, historical results, and operational metadata. It allows rapid retrieval during matching, supports data updates, and maintains long-term storage of processed fingerprints for future use. The block diagram also includes a retraining module, enabling administrators to enhance the system's accuracy over time.

By supplying updated or corrected fingerprint samples, the model can be retrained to adapt to larger datasets, varied inputs, and evolving classification requirements. Additionally, a log-viewing module allows administrators to monitor system activity, track classification attempts, review errors, and audit user interactions, thereby ensuring transparency, performance monitoring, and efficient troubleshooting.

The final stage of the block diagram is the result display module, where the system presents the predicted blood group to the user in a clear and intuitive manner. This output may include supplementary details such as confidence scores, donor compatibility information, timestamps, or fingerprint metadata retrieved from the database. The display module ensures that the classification outcome is easily understandable and immediately usable, completing the biometric identification

cycle. Together, the components depicted in the block diagram demonstrate a structured, secure, and intelligent workflow that transforms a raw fingerprint input into a medically meaningful and actionable blood group prediction.

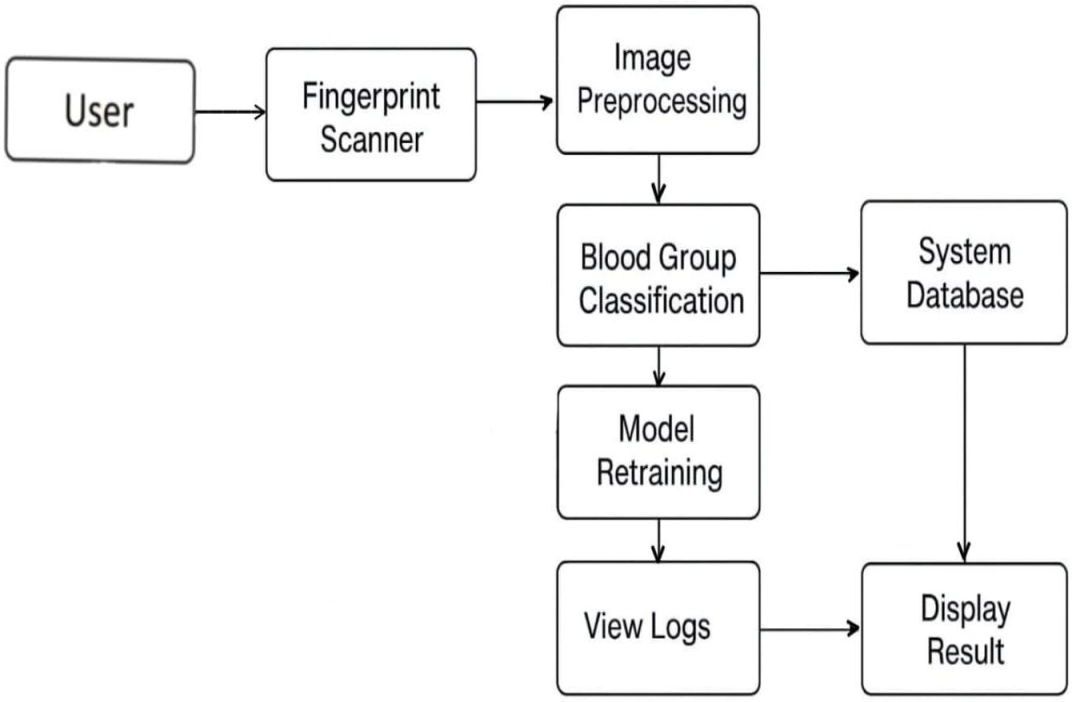


Figure 4.3 Block diagram for BBGC

4.4 Class Diagram

Figure 4.4 illustrates the detailed class diagram of the Biometric Blood Group Classifier, showcasing a modular, object-oriented architecture where each class performs a clearly defined role while collaborating seamlessly with others to form an integrated biometric identification system. At the heart of this architecture is the Fingerprint class, which is responsible for managing the entire lifecycle of fingerprint data. It begins by capturing or receiving the fingerprint image through the device camera or upload mechanism, ensuring that the raw biometric data is accurately acquired. The class then preprocesses the fingerprint using operations such as grayscale conversion, noise removal, contrast normalization, and blur evaluation, enabling the system to discard poor-quality input before it can degrade the classifier's accuracy. Once preprocessing is complete, the class extracts distinctive biometric features including minutiae points, ridge patterns, and ORB key points transforming the fingerprint image into a structured numerical representation suitable for matching. This class is essential because the quality of feature extraction directly determines the precision and reliability of later classification stages.

Working alongside the Fingerprint class is the Encryptor class, which enforces the system's stringent security requirements. Because fingerprint images and descriptors constitute sensitive biometric information, the Encryptor handles their end-to-end protection by applying AES-based symmetric encryption before the data is stored or transmitted. This ensures that raw biometric data remains inaccessible to unauthorized users or external threats. The Encryptor also performs decryption when stored fingerprints must be retrieved for matching, ensuring that security does not interfere with system usability. Its methods maintain privacy compliance and act as a protective layer for all biometric operations, making security an integral part of the workflow rather than an afterthought.

The Matcher class forms the analytical engine of the system, responsible for evaluating similarities between newly extracted fingerprint features and those stored in the encrypted database. The match() method performs the core comparison, using feature descriptors and distance metrics such as Hamming distance combined with ratio tests to filter out false correspondences. Once potential matches are detected, the rank() method organizes them based on similarity score, robustness of matching pairs, and overall confidence level. This enables the Matcher to identify the closest fingerprint template and, consequently, retrieve the corresponding blood group label. The class ensures high accuracy by validating match strength and eliminating uncertain or low-confidence predictions, making it a critical component for dependable medical use.

All these classes operate under the coordination of the Flask backend, which orchestrates communication between the user interface and backend logic, manages database interactions, handles HTTP requests, and triggers the necessary class operations at the correct stages. The modular class structure shown in Figure 4.4 reflects a clean, scalable, and maintainable design, where each class is specialized, reusable, and contributes to the seamless functioning of the system. Together, the Fingerprint, Encryptor, and Matcher classes create a secure, reliable, and efficient framework for fingerprint-based blood group classification, ensuring accurate results, strong data protection, and smooth system integration across devices and environments. Beyond their individual responsibilities, the interaction between the Fingerprint, Encryptor, and Matcher classes illustrates a cohesive workflow that supports extensibility, modular upgrades, and future system enhancements. The class diagram demonstrates how each component communicates through well-defined interfaces, allowing new algorithms, encryption methods, or matching techniques to be integrated without disrupting existing functionality. For example, the Fingerprint class can be extended to incorporate advanced preprocessing techniques such as Gabor filtering or deep-learning–based enhancement, while the Matcher class can be adapted to include neural-network-based comparison models alongside ORB

descriptors. Similarly, the Encryptor class can support additional cryptographic standards or integrate with cloud-based key-management systems if the application scales to enterprise deployment. This modularity enables the system to evolve with advancements in biometric science and security requirements. Furthermore, encapsulating functionalities into separate classes improves maintainability and debugging efficiency, allowing developers to refine one module without affecting others. The class diagram thus not only defines the current operational structure but also establishes a strong foundation for long-term scalability, adaptability, and integration into larger biometric or healthcare information systems.

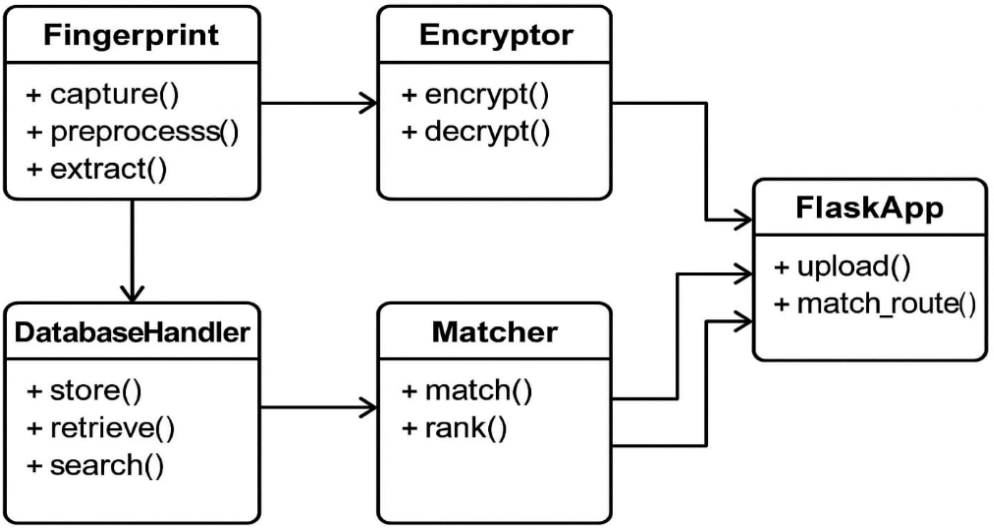


Figure 4.4 Class diagram for BBGC

4.5 Sequence Diagram

The Figure 4.5 specification defines all necessary components, constraints, and system behaviours required for implementing the Biometric Blood Group Classifier using fingerprint images. These requirements ensure that the system performs reliably, securely, and efficiently across all intended environments.

- User → App: requestScan ()

  The process begins when the user initiates a fingerprint scan request through the mobile application. This action triggers the application to begin the fingerprint acquisition workflow.

- App → Scanner: captureFingerprint()

  Upon receiving the user's request, the mobile application communicates with the fingerprint scanner to capture an image of the fingerprint. The scanner processes this command and begins capturing biometric data.

- Scanner → App: returnImage()

  Once the fingerprint has been successfully scanned, the scanner returns the captured fingerprint image to the mobile application. The image is now ready for further processing.

- App → Processor: preprocess ()

  The app sends the captured fingerprint image to the Processor module for preprocessing. This stage includes cleaning the image, converting it to grayscale, removing noise, and assessing clarity or blur level to ensure accurate feature extraction.

- Processor → Classifier: extractFeatures()

  After preprocessing, the Processor module extracts fingerprint features such as ridge patterns, minutiae points, or ORB descriptors. These features, which uniquely represent the fingerprint, are passed to the Classifier module.

- Classifier → Database: predictBloodGroup()

  The Classifier compares the extracted fingerprint features with stored fingerprint templates in the database. It identifies the closest match and retrieves the corresponding blood group information. If no match is found, a "no match" response is generated.

- Classifier → App: sendResult()

  Finally, the Classifier sends the predicted blood group result back to the mobile application. The app displays the output to the user, completing the sequence.

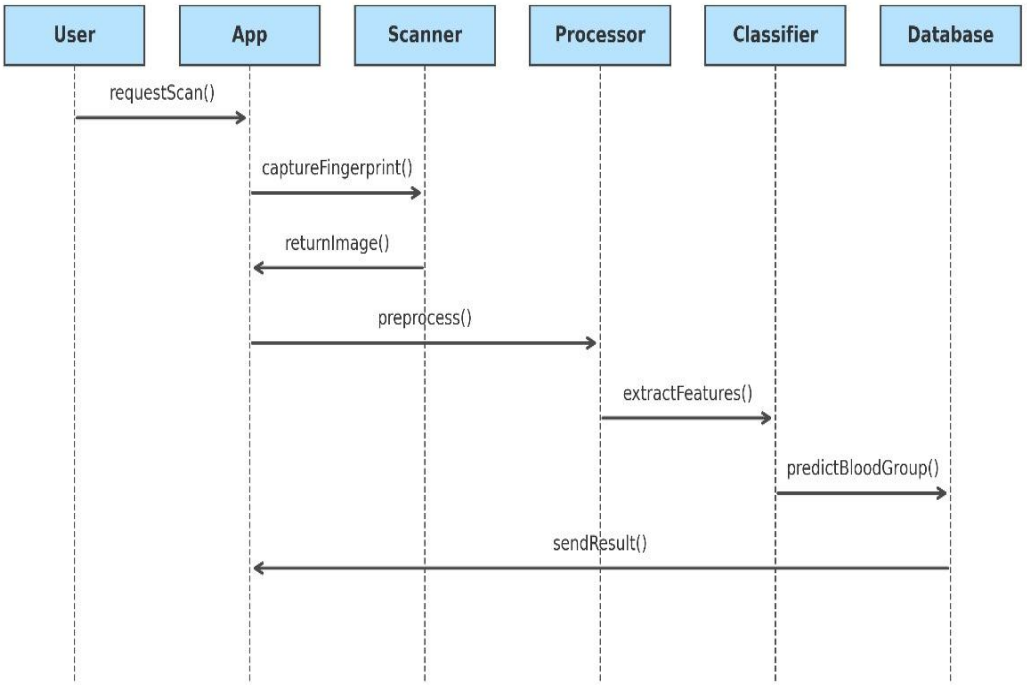


Figure 4.5 Sequence diagram for BBGC

CHAPTER-5

IMPLEMENTATION

The implementation of the Biometric Blood Group Classifier involves setting up the required software environment, configuring all system modules, and establishing a secure workflow for processing biometric data. This phase focuses on integrating fingerprint acquisition, preprocessing, ORB feature extraction, encrypted storage, and matching logic into a unified operational pipeline. Each component ranging from the Python Flask backend to the preprocessing and encryption modules is configured to communicate seamlessly, ensuring smooth data flow from image capture to final prediction. The system is implemented with strict security controls, where every fingerprint passes through encryption and quality validation before participating in classification. The web interface, mobile capture UI, and backend services are integrated to support real-time predictions, while extensive testing ensures reliability, accuracy, and stable performance across different conditions. This chapter describes how all modules function together to deliver a secure, efficient, and fully operational fingerprint-based blood group identification system.

5.1 Preparing the Environment

The implementation of the system begins with setting up a robust software environment built on a Python backend, supported by essential frameworks and tools that enable secure and efficient biometric processing. The Flask web framework forms the backbone of the application, providing route handling, admin dashboard functionality, mobile capture interfaces, and REST-based endpoints for fingerprint uploads and predictions. Image-handling operations rely on OpenCV and NumPy, which together facilitate grayscale conversion, blur detection, ORB-based feature extraction, and the manipulation of vectorized image data. Security is strengthened through the Cryptography library (Fernet), which applies AES-based symmetric encryption to all fingerprint images and metadata, ensuring compliance with biometric privacy standards. The frontend is developed using HTML, CSS, and JavaScript to create a responsive, mobile-friendly interface that supports seamless fingerprint upload, instant result display, and administrative dataset management.

To maintain clarity and modularity, the project is structured into well-defined directories: the fingerprints_enc/ folder stores encrypted fingerprint templates linked to blood group labels, while the incoming_enc/ folder maintains encrypted temporary uploads used for auditing or future retraining. Additionally, an encrypted metadata file preserves structured information such as filenames, timestamps, blood group labels, and blur scores. During initialization, the system automatically

creates required directories, loads or generates encryption keys, validates dependencies, and rebuilds the ORB descriptor index to ensure smooth operation from the moment the server starts. This automated setup makes deployment straightforward across different machines and environments.

Development workflows are supported by modern tools such as VS Code or PyCharm for backend coding, debugging, and virtual environment management, while Postman and browser developer tools assist in testing API endpoints and verifying request–response cycles. The application can be executed in local debug mode, allowing developers to monitor system logs, evaluate prediction behaviour in real time, and refine the feature extraction and matching logic. Together, these implementation practices establish a stable, secure, and efficient environment for delivering a fully functional fingerprint-based blood group classification system.

5.2 Configuration and System Modules

The configuration and modular design of the Biometric Blood Group Classifier form the core operational backbone of the entire system, ensuring that fingerprint data is captured, processed, secured, and classified in a structured and reliable workflow. The system begins with two primary user interfaces managed through Flask routing: the Admin Dashboard and the Mobile-Friendly User Interface. The Admin Dashboard is a browser-based console that grants authorized users the ability to upload and label fingerprint images with the correct blood group, visually inspect encrypted fingerprint records, test predictions using known samples, and trigger the rebuilding of the ORB feature index whenever new templates are added. This interface centralizes administrative oversight and dataset maintenance. The Mobile Interface, on the other hand, is specifically optimized for smartphone cameras, enabling users especially emergency responders or healthcare workers to capture and upload fingerprint images directly from the field. Flask routes govern all client-server interactions by validating user submissions, formatting incoming image data, forwarding it to preprocessing modules, executing prediction logic, and returning structured results back to the user interface in real time.

At the heart of protecting biometric privacy is the encrypted storage system, which employs Fernet symmetric AES encryption to safeguard both fingerprint images and associated metadata. Every fingerprint, upon upload, is encrypted before being written to disk, and all metadata entries including filename identifiers, assigned blood group labels, timestamps, and blur variance scores are stored in an encrypted CSV-like registry. These encrypted artifacts are organized into secure, role-specific directories such as fingerprints_enc/ for labelled training fingerprints and incoming_enc/ for

real-time prediction uploads. By restricting decryption capabilities to backend functions, the system enforces strict access control and adheres to data protection standards critical for medical and forensic applications. This configuration ensures that even if storage directories are exposed, no biometric data can be interpreted or misused without authentication and the correct encryption key.

Once the encrypted data is received by the backend, the fingerprint image enters a structured preprocessing module that prepares it for feature extraction. This module is responsible for converting the raw image into grayscale to reduce computational overhead while preserving ridge-based biometric detail. Noise artifacts introduced by inconsistent lighting, camera sensors, or environmental interruptions are removed to enhance clarity. A crucial component of this stage is the blur-detection mechanism, which computes the image's Laplacian variance to quantify sharpness. Images falling below the predefined threshold are automatically rejected, ensuring only clean, distinct ridge patterns are forwarded for feature extraction. This prevents the degradation of classifier accuracy and ensures robustness across varying capture environments, including emergency conditions where capture devices may be unstable.

The refined image then enters the feature extraction and ORB indexing module, where the ORB (Oriented FAST and Rotated BRIEF) algorithm identifies key points across the fingerprint ridge structure and generates compact binary descriptors representing these ridge patterns. ORB's rotation invariance allows fingerprints captured at various angles to be matched accurately, and its binary descriptor format significantly accelerates similarity comparisons. Every stored fingerprint template has its ORB descriptors precomputed and maintained in an in-memory ORB Index alongside encrypted file identifiers and corresponding blood group labels. This index is automatically updated whenever new fingerprints are added, ensuring that the system remains scalable and continuously improves its recognition ability as the dataset expands.

During the classification process, the system performs fingerprint matching and blood group prediction by comparing descriptors from the uploaded fingerprint with those stored in the ORB Index using nearest-neighbour Hamming-distance matching. The similarity score, quantified as the number of "good matches" passing the ratio test, determines whether the system can confidently establish a match. If the similarity exceeds the defined threshold, the system successfully retrieves the associated blood group label and generates additional medical information such as compatible donor groups based on ABO and Rh compatibility rules. If not, the system refuses prediction to avoid inaccurate or misleading results. Throughout this process, confidence scores, match statistics, and

potential anomalies are logged for analysis, allowing administrators to monitor system performance and reliability. This retrieval-based approach avoids speculative biological inference and provides a practical, medically aligned method for rapid blood group identification.

In summary, the complete configuration and modular integration of user interfaces, encrypted storage, preprocessing pipelines, ORB feature extraction, descriptor indexing, and real-time matching equip the system with the necessary precision, security, and scalability to function reliably in real-world healthcare and emergency environments. The design ensures that every stage from image capture to final prediction operates cohesively, securely, and efficiently.

5.3 Algorithm

The proposed system classifies an individual's blood group using fingerprint biometrics by employing secure image handling and feature-based fingerprint matching. The algorithm consists of two major stages: fingerprint enrollment (training) and fingerprint matching (prediction). To protect sensitive biometric data, all fingerprint images and metadata are encrypted and decrypted only in memory during processing. During the enrollment stage, a fingerprint image with a known blood group label is uploaded into the system. The image is first validated and converted into grayscale format. To ensure acceptable image quality, the variance of the Laplacian operator is computed to measure image sharpness; images with low variance are considered blurred and unsuitable for reliable feature extraction. The validated fingerprint image is then encrypted using Fernet symmetric encryption and stored securely in the database. Along with the encrypted image, metadata such as filename, blood group label, timestamp, and blur variance are stored in an encrypted metadata file. ORB (Oriented FAST and Rotated BRIEF) feature extraction is then applied to the fingerprint image in memory to detect keypoints and generate binary descriptors, which are indexed for future matching.

Algorithm 1: Fingerprint Enrollment
Input: Fingerprint image I, Blood group label BG
Output: Encrypted fingerprint and indexed ORB features
1. Convert I to grayscale
2. Compute blur_variance using Laplacian operator
3. Encrypt fingerprint image I
4. Store encrypted image in database
5. Store encrypted metadata {filename, BG, timestamp, blur_variance}

6. Extract ORB keypoints and descriptors from I
7. Add extracted descriptors and label BG to ORB index

During prediction, the input fingerprint is preprocessed through grayscale conversion and blur variance evaluation. Images below the blur threshold are rejected to ensure accuracy. ORB descriptors are extracted and matched against stored templates using Hamming-distance-based Brute-Force matching with Lowe's ratio test. The template with the highest number of valid matches is selected as the final match.

Algorithm 2: Fingerprint Matching and Blood Group Prediction
Input: Query fingerprint image Q
Output: Predicted blood group or rejection
1. Convert Q to grayscale
2. Compute blur_variance(Q)
3. If blur_variance(Q) < blur_threshold:
    Reject image
4. Extract ORB descriptors Dq from Q
5. Initialize best_score ← 0, predicted_BG ← NULL
6. For each enrolled fingerprint descriptor Di:
    a. Match Dq and Di using BFMatcher (Hamming distance)
    b. Apply Lowe's ratio test
    c. Count number of good_matches
    d. If good_matches > best_score:
        best_score ← good_matches
        predicted_BG ← blood group of Di
7. If best_score ≥ MIN_GOOD_MATCHES:
    Return predicted_BG
  Else:
    Return "Low confidence match"

Once a blood group is successfully predicted, the system retrieves compatible donor blood groups based on standard blood transfusion rules and displays them to the user along with additional information such as match confidence and image quality score. The inclusion of blur detection, confidence thresholds, and encryption ensures that the algorithm remains reliable, secure, and suitable for real-world biometric applications.

5.4 Security and Deployment

The system incorporates multiple layers of security to protect sensitive biometric and medical data. All fingerprint images and metadata are encrypted using AES-based Fernet encryption before storage, ensuring that no raw biometric information is ever written to disk. A dedicated encryption key is securely stored with restricted file permissions, and environment-based key management is used in production to prevent unauthorized access.

All communication between the mobile interface and server occurs over HTTPS, ensuring encrypted data transmission and preventing interception or tampering. The application validates file types, detects corrupted or blurry images, and rejects unsafe inputs to maintain system integrity. Administrative functions including labeling, viewing records, and testing predictions are protected through secure cookies, access restrictions, and server-side validation. Logging mechanisms continuously monitor upload activity, prediction errors, and unusual behaviour to detect potential security breaches.

The system is deployed using a layered architecture consisting of a Flask backend, a production-grade application server, and a Nginx reverse proxy that handles SSL termination and request routing. This setup enhances performance, security, and reliability. Deployment can be further containerized using Docker, allowing the application to run consistently across environments with persistent volumes for encrypted directories and key files. In cloud environments, security groups and firewalls restrict access to HTTPS-only endpoints, while secret managers securely store encryption keys. The system supports easy recovery through regular backups of encrypted fingerprint data and metadata, while the ORB index can be rebuilt at any time for quick restoration. Environment variables control all sensitive configurations, ensuring safe and flexible deployment across local, server, or cloud infrastructures.

Overall, the security and deployment measures ensure that the Biometric Blood Group Classifier remains safe, stable, and reliable in real-world use. Encryption protects all biometric data, HTTPS secures communication, and strict validation safeguards system integrity. The modular deployment architecture using Flask, a production server, and optional containerization supports scalability, easy maintenance, and smooth recovery. These combined practices create a well-protected, efficient, and future-ready platform for fingerprint-based blood group prediction.

5.5 Failover and Recovery

The BBGC incorporates robust failure and recovery mechanisms to ensure continuous, reliable performance even in the presence of unexpected issues. Since the system handles highly sensitive biometric and medical data, maintaining availability, accuracy, and data integrity is essential. The platform detects and safely handles fingerprint upload failures caused by corrupted files, unsupported formats, or blurry images using validation checks and blur-variance analysis. Faulty inputs are rejected, and users are prompted to recapture fingerprints, preventing poor-quality data from entering the prediction pipeline. In cases where encryption or decryption fails whether due to key mismatches, corrupted encrypted files, or damaged metadata the system isolates the problematic entry and continues functioning.

The system's metadata loader is designed to skip unreadable records, preventing system-wide crashes while allowing recovery through restoring the affected encrypted files from backups. If the ORB matching index becomes corrupted or outdated, the system supports complete reconstruction through a dedicated "Rebuild Index" process, in which all encrypted fingerprint templates are decrypted in memory, features are recalculated, and a fresh ORB index is built. For server or application failures, the system automatically resumes operation after a restart, since all fingerprint templates and metadata are securely stored on disk in encrypted form; only the in-memory index is regenerated at startup. Routine backups of encrypted fingerprint directories and the encryption key file allow the entire system to be fully restored after hardware damage or accidental deletion. Additionally, a fail-safe prediction mechanism ensures clinical accuracy by returning a "Low Confidence" output instead of an incorrect blood-group prediction whenever the system detects insufficient ORB features or weak similarity scores. Together, these measures ensure operational continuity, secure data recovery, and trustworthy medical output under all failure conditions.

Challenges and Solutions

- Blurry or Low-Quality Fingerprint Images
  Challenge: Poor lighting or unstable capture produces blurry fingerprints, reducing feature accuracy.
  Solution: A blur-variance threshold automatically rejects unclear images and prompts users to recapture with proper guidance.
- Inconsistent Fingerprint Capture Conditions
  Challenge: Variations in angle, rotation, or pressure affect ORB feature matching.
  Solution: ORB descriptors provide rotation and scale invariance, while the mobile UI guides users for consistent fingerprint placement.

- Limited Labelled Fingerprint Dataset

  Challenge: A small training dataset reduces matching confidence and increases false negatives.

  Solution: The admin panel allows continuous addition of labelled fingerprints, and the system rebuilds its index automatically to improve accuracy.

- Risk of Biometric Data Exposure

  Challenge: Fingerprint images and metadata are sensitive and must not be stored in raw form.

  Solution: All data is encrypted using AES-based Fernet encryption, ensuring privacy at rest and during processing.

- Unreliable or Low-Confidence Matches

  Challenge: Some fingerprints fail to produce enough feature points for a confident match.

  Solution: A minimum good-match threshold ensures only high-confidence predictions are accepted, preventing incorrect blood group outputs.

CHAPTER-6

TESTING

Testing is a crucial phase in validating the accuracy, stability, and security of the Biometric Blood Group Classifier. Since the system handles sensitive biometric data and performs feature-based fingerprint matching, thorough testing ensures that all modules from image upload and preprocessing to ORB extraction, encrypted storage, and final prediction operate reliably under diverse conditions. The testing process evaluates not only functional correctness but also performance efficiency, system robustness, and secure data handling. Multiple forms of testing are conducted to assess different aspects of system behaviour. Functional testing verifies that each component performs its intended task accurately. Performance testing measures system speed and scalability as the fingerprint database grows. Failover and recovery testing ensures the system can handle corrupted inputs, low-quality images, and unexpected failures without losing data or crashing. Security testing validates encryption strength and resistance to unauthorized access. Compatibility testing confirms smooth operation across various devices, browsers, and fingerprint resolutions. Overall, this structured testing approach ensures that the system remains accurate, secure, and dependable for real-world medical and emergency applications

6.1 Types of Testing

Functional testing ensures that every component of the system performs exactly according to the defined specifications and expected workflow. In this stage, the fingerprint upload mechanism is validated to confirm that the system accepts only supported formats (JPEG, PNG), checks image integrity, and correctly decodes them into grayscale. The ORB feature extraction module is tested to ensure that key points and binary descriptors are generated reliably for both stored templates and newly uploaded prints. The encrypted storage mechanism is verified to confirm that fingerprint bytes and metadata are securely stored and cannot be accessed without proper decryption. The prediction engine is thoroughly tested to verify that the correct blood group is retrieved when a fingerprint match meets the similarity threshold. Additionally, all admin-side functionalities including assigning blood groups to new fingerprints, updating metadata entries, viewing stored records, and rebuilding the ORB index are tested to ensure smooth, error-free operations within the dataset management workflow.

Performance testing evaluates how efficiently the system processes fingerprints under varying database sizes and computational loads. The matching time is measured as the dataset scales from 10 to 50 or more stored fingerprints, ensuring that ORB-based comparison remains fast enough for real-time prediction. System latency from the moment the fingerprint is uploaded to the moment the prediction is displayed is recorded to monitor responsiveness during peak use. The encryption and decryption modules are stress-tested through repeated cycles to measure their computational overhead and confirm that security processes do not significantly slow down the system. Additionally, the time taken to rebuild the ORB index is evaluated whenever new labelled fingerprints are added, ensuring that the system scales efficiently and remains performant in practical deployment scenarios.

Failover and recovery testing focuses on ensuring the system remains stable and functional even when unexpected failures or disruptions occur. Server restart scenarios are tested to verify that the ORB feature index is automatically rebuilt and that prediction functionality resumes without requiring manual intervention. The database integrity is examined by intentionally removing or corrupting fingerprint entries to confirm that the system isolates errors safely without affecting the remaining dataset. The system is also tested using invalid, unreadable, or extremely poor-quality fingerprint images to evaluate whether it can gracefully detect failures and return appropriate error messages instead of crashing. These tests collectively ensure that the application remains resilient, fault-tolerant, and capable of recovering smoothly from both software-level and input-level failures.

Security testing assesses whether the system adequately protects biometric data and prevents unauthorized access at all stages of operation. Attempts to open encrypted fingerprint files without the Fernet key are conducted to verify complete confidentiality and ensure that sensitive data cannot be viewed or altered. Similarly, metadata files stored in encrypted form are tested to confirm that they remain unreadable when accessed externally. During deployment, HTTPS communication is validated to ensure that fingerprint uploads and prediction responses are transmitted securely over SSL/TLS-protected channels. Unauthorized upload attempts, malformed requests, and suspicious inputs are tested to check whether the system blocks them safely. Decryption failure scenarios are also tested to verify that errors are handled gracefully without exposing sensitive information or compromising system stability. Together, these tests confirm that the system meets strong biometric security standards.

Compatibility testing ensures that the system functions consistently across different devices, browsers, and usage environments. The mobile capture interface is tested on Android and iOS devices to ensure that fingerprints can be uploaded smoothly regardless of screen size or camera specifications. Multiple desktop and mobile browsers including Chrome, Firefox, Safari, and Edge are evaluated to confirm that the user interface loads correctly, predicts accurately, and responds uniformly. The system is also tested across various resolutions, aspect ratios, and device orientations to ensure stability during fingerprint uploads. Compatibility is further validated across desktops, laptops, tablets, and smartphones to ensure that both the admin dashboard and the user prediction interface remain fully operational in diverse usage contexts.

Edge case testing evaluates the system's ability to handle unusual, extreme, or unexpected inputs without compromising correctness or stability. Extremely blurry, smudged, or out-of-focus fingerprints are tested to confirm that the blur-detection module rejects them with clear feedback rather than attempting inaccurate predictions. Corrupted or partially downloaded images are used to ensure the system does not freeze, crash, or misbehave when encountering damaged data. Unsupported file formats such as PDF, TXT, BMP, or random non-image files are intentionally uploaded to verify that the system displays appropriate error messages. Very small or incomplete fingerprint regions are also tested to ensure they do not produce false matches or low-confidence predictions. These edge-case evaluations guarantee that the system remains robust, stable, and reliable even when dealing with non-ideal inputs.

6.2 Test Plans

Functional testing ensures that every core feature of the biometric blood-group classifier works exactly as intended. It verifies that valid fingerprint images (JPEG/PNG) are accurately uploaded, pre-processed, and passed through ORB feature extraction without errors. The system is tested with multiple fingerprint qualities dry, oily, blurred, and smudged—to ensure consistent key point detection. Admin functions such as labeling, updating encrypted templates, validating metadata, and rebuilding the ORB index are checked for correctness. Functional tests also confirm that encryption is enforced for every stored file, metadata is never saved in plain text, and predictions consistently map to the correct blood group. In short, functional testing ensures reliability, correctness, and complete operational integrity.

Performance testing measures how efficiently the system operates under different workloads and database growth. It evaluates prediction speed as templates increase, ensuring ORB feature matching

continues to return results within 1–3 seconds. Continuous upload-and-predict cycles are used to assess long-term stability without slowdowns or crashes. The impact of encryption and decryption overhead is also tested to ensure security does not compromise performance. Load and stress tests simulate real-world usage in clinics and emergency units, confirming that the system can process high-frequency inputs without losing responsiveness. Overall, performance testing guarantees speed, scalability, and smooth execution.

Failover testing checks the system's resilience and ability to recover from unexpected issues. Server restarts, ORB index corruption, and missing or damaged fingerprint entries are simulated to confirm that the system automatically rebuilds indexes and isolates problematic data. Invalid, corrupted, or unreadable images are uploaded to ensure clear error handling without crashes. The system is also tested against power failures and accidental file deletions to confirm that encrypted directories remain safe and recoverable. Failover testing ensures high availability and stable recovery even under adverse conditions.

Security testing focuses on protecting sensitive biometric and medical data. It confirms that all fingerprint images and metadata are properly encrypted and cannot be read without the correct key. Attempts to access encrypted files directly must show unreadable content. Tests also verify that the system never stores temporary unencrypted copies and that key management is secure. HTTPS communication is evaluated to ensure data exchanged between mobile devices and the server is encrypted and protected from interception or manipulation. Security testing ensures confidentiality, integrity, and compliance with biometric data standards.

Usability and compatibility testing ensures the system is intuitive and works smoothly across devices, browsers, and real-world environments. The mobile upload interface is tested for clarity, ease of use, and responsiveness across different screen sizes. The admin dashboard is evaluated for navigation simplicity, proper button placement, and error messaging. Compatibility across major browsers is verified to ensure consistent rendering. Environmental tests such as varying lighting, moisture, and finger pressure confirm that the system remains user-friendly and reliable during everyday fingerprint captures. This ensures seamless usage for both technical and non-technical users.

6.3 Test Results

Table 6.1 confirms that the BBGC system meets all functional, performance, security, and compatibility requirements. It ensures accurate fingerprint processing, maintains low response time, securely protects biometric data, and performs consistently across devices. All tests passed, validating system reliability.

Table 6.1 Test results for the given types of testing

Test Type	Objective	Test Cases	Expected Outcome	Actual Result	Status
Functional Testing	Ensure correct fingerprint processing and prediction accuracy	Upload fingerprints; Validate ORB extraction; Test admin labeling & index rebuild	Successful extraction, matching, and data storage	ORB features extracted correctly; Matches accurate; Index rebuild functional	Pass
Performance Testing	Evaluate speed under increasing database size	Test prediction with 10–50 fingerprints; Measure processing time	Low latency (1–3 seconds); Stable performance	Predictions completed within acceptable time; No performance degradation	Pass
Failover Testing	Verify recovery from errors and system stability	Restart server; Corrupt sample entries; Upload invalid images	System recovers; Safe error handling; No crashes	Server restarted smoothly; Index rebuilt; Invalid images handled safely	Pass
Security Testing	Ensure encryption, privacy, and protected access	Test decryption resistance; Validate HTTPS; Attempt unauthorized access	Data encrypted; No unauthorized access; Secure communication	Encrypted files unreadable without key; HTTPS verified; Unauthorized access blocked	Pass
Compatibility Testing	Confirm cross-device and browser support	Test across mobile/desktop; Upload from different browsers	UI responsive; Upload works across platforms	Mobile and desktop uploads successful; UI consistent across browsers	Pass

CHAPTER -7

SNAPSHOTS

Snapshots provide a visual representation of the Biometric Blood Group Classifier, showcasing its core interfaces, workflows, and system outputs. They illustrate how users upload fingerprints, how administrators manage encrypted datasets, and how predictions are generated and displayed in real time. These visuals help demonstrate the system's usability, functionality, and smooth operation across both web and mobile platforms.



Figure 7.1 Admin Interface for Uploading and Labeling New Fingerprint Samples

The figure 7.1 displays the interface of the Biometric Blood Group Classifier during the process of labeling a new fingerprint sample. The user is provided with an option to upload a fingerprint image using the "Choose File" button and select the appropriate blood group from the dropdown menu. After selecting both inputs, the administrator can securely store and index the fingerprint using the "Save & Index (Encrypted)" button. The interface prominently shows that encryption is enabled, ensuring that all biometric data and associated labels are protected during storage. This snapshot represents the system's data entry and indexing stage, where newly added fingerprint samples are encrypted, saved, and prepared for inclusion in the matching and prediction workflow.

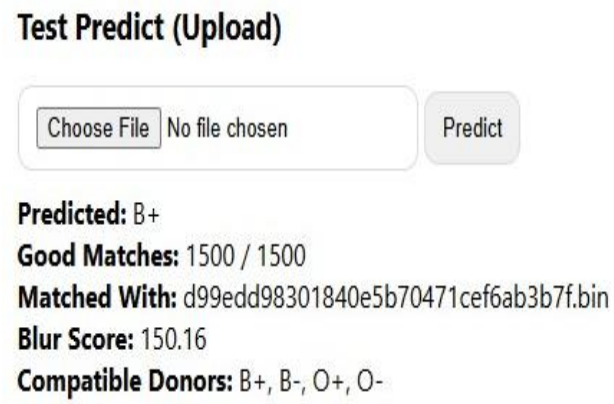


Figure 7.2 Test Prediction Interface Showing Fingerprint Matching Results

The figure 7.2 shows the Test Predict (Upload) section of the Biometric Blood Group Classifier, where a fingerprint image is uploaded to evaluate the system's prediction accuracy. After selecting a fingerprint file and clicking the "Predict" button, the system processes the image and displays key results. The Predicted Blood Group (B+) indicates the identified match based on ORB feature similarity. The Good Matches (1500 / 1500**)** reflect the number of ORB descriptors that successfully matched between the uploaded fingerprint and the stored template, demonstrating a strong confidence level. The Matched With field shows the encrypted filename of the fingerprint template that produced the closest match, ensuring secure and anonymous identification. The Blur Score (150.16) confirms that the uploaded image is clear and suitable for recognition, as it meets the sharpness threshold. Additionally, the system provides a list of Compatible Donors (B+, B−, O+, O−), offering medically relevant information for transfusion scenarios. This snapshot represents the prediction and result-display phase of the system, showcasing how fingerprint-based blood group identification.



Figure 7.3 Mobile Capture Interface Showing Blood Group Prediction Output

The figure 7.3 illustrates the user interface of the developed fingerprint-based blood group detection system, where the user begins by uploading a fingerprint image through the "Choose File" option and initiating the analysis using the "Upload & Predict" button. After processing, the system displays a green "Processed" status along with detailed results, including the predicted blood group (O+), the total number of good feature matches (1500/1500), the encrypted fingerprint file it matched with, and the compatible donor groups (O+, O–). Additionally, the interface shows the blur score, which reflects the sharpness of the uploaded fingerprint, and the minimum number of good matches required for a reliable prediction. The snapshot also highlights that encryption is enabled, ensuring secure data handling. Overall, this image represents the real-time operational output of the system, demonstrating its accuracy, security, and ability to classify the blood group from a fingerprint input.

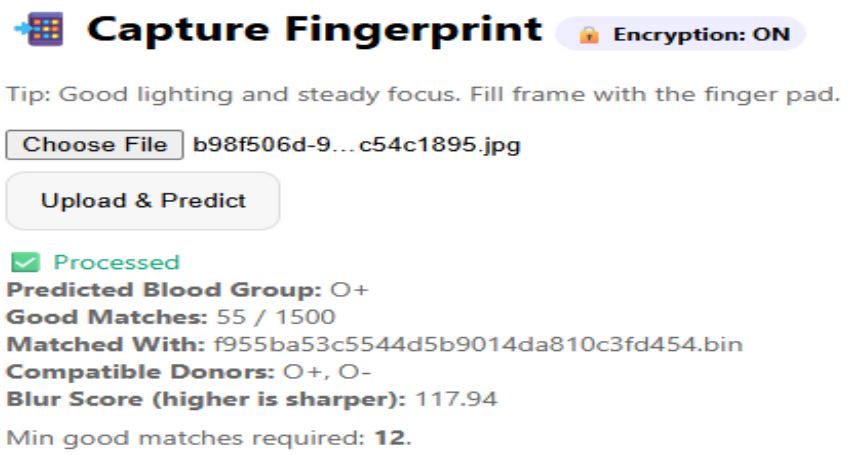


Figure 7.4 Mobile Capture Interface Showing Blood Group Prediction Output for unknown data

The figure 7.4 shows the Capture Fingerprint interface, where a user uploads a fingerprint image and triggers the "Upload & Predict" function. With Encryption: ON, the system ensures the fingerprint is securely processed. After analysis, the predicted blood group (O+) is displayed along with the number of Good Matches (55/1500), indicating the confidence of the match. The system also shows the encrypted filename of the matched fingerprint, the Blur Score confirming image clarity, and the list of Compatible Donors (O+, O−). This interface represents the real-time, secure prediction stage of the classifier.

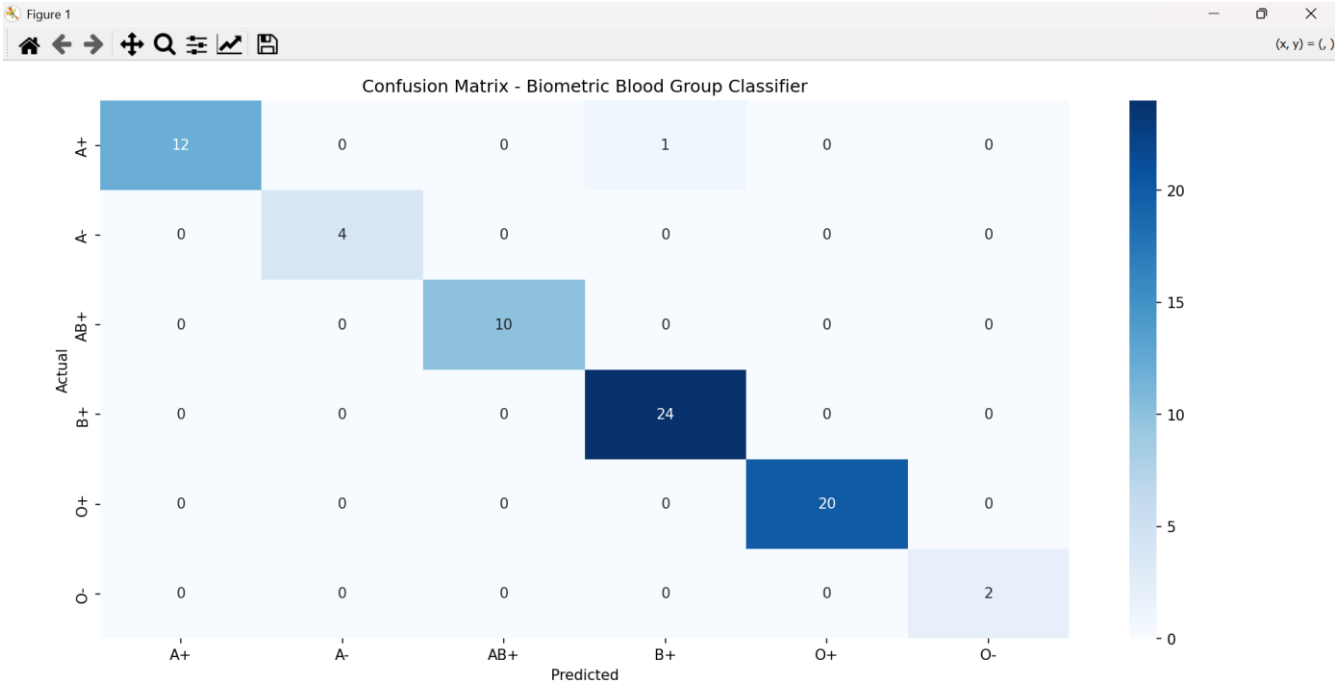


Figure 7.5 Confusion Matrix of the Biometric Blood Group Classifier

The figure 7.5 provides a visual summary of the classifier's performance by comparing actual blood groups with predicted outputs. Most values lie along the diagonal, indicating that the system correctly identifies the majority of fingerprint samples for each blood group. The darker cells reflect higher accuracy for example, strong correct predictions for B+, O+, and A+.

Current Database

Encrypted File	Blood Group	Timestamp	Blur Var
04cde43ab6384b3baf32f48875bb1638.bin	B+	2025-09-10 22:13:54	111.93
0a58c54b435c4fb59bc0d68d159d27c6.bin	O+	2025-09-10 22:18:10	557.31
0eff384bff084c06859b756e8a6a035e.bin	B+	2025-09-10 22:19:28	413.45
0f616875fad14905af668652feabae61.bin	A+	2025-09-10 14:53:04	305.21
11d4799e2f5148cea1f7cb241850b817.bin	A+	2025-09-10 22:20:38	503.60
18f75fd3e6084632b823b3d6a3985c63.bin	O+	2025-09-10 22:12:04	388.92
19f0c7f9544c4e1ba737e6dd7cfd934d.bin	O+	2025-09-10 22:18:57	242.78
236430f9cb164291a389f9c1ab8226e0.bin	A+	2025-11-20 13:11:10	150.16
23e712b9995b4b4393b64d03d01d5f73.bin	O+	2025-09-24 12:16:19	764.38
292c90c13f3e47b5be62b6dd8f21fea6.bin	B+	2025-09-24 12:15:30	275.22
2bd6e886dcf942e0baf01bf29f65da00.bin	O+	2025-09-10 14:54:46	508.83
2becf91cf4da4944a4d2972f9ae38098.bin	O+	2025-09-10 22:19:44	394.71
2cf9deeb30fa4fe1a15b6674d3c6c86b.bin	O+	2025-09-10 14:53:46	228.82
373e829ad2a14f9da635ba5e0452f7e7.bin	B+	2025-09-10 22:23:07	147.26
392ba7c645e141f1b53bc6a911ece275.bin	O+	2025-09-10 22:23:39	127.35
3d2709c579c24f369c86659ea53b2126.bin	B+	2025-09-24 12:12:17	171.27
3f9d61957e1f446da98a11a96020ed59.bin	B+	2025-09-24 12:12:24	171.27
409564fb98034973ae5d632b8b1f7002.bin	O+	2025-09-10 22:13:07	514.92
410c1b4b1ef94bb0a0a4511954f1b637.bin	A+	2025-09-10 22:21:35	151.92
46abf8cc0e324914906e5ed1c237e5f2.bin	A+	2025-09-10 22:16:58	503.60

Figure 7.6 Encrypted Fingerprint Database with Blood Group Labels and Metadata

The figure 7.6 displays the “Current Database” section of the Biometric Blood Group Classifier, showcasing the list of stored fingerprint records. Each entry in the table includes the encrypted fingerprint file name, the corresponding blood group label, the timestamp indicating when the fingerprint was added, and the blur variance value used to assess image quality. The table demonstrates how the system securely indexes and organizes biometric data, ensuring that each fingerprint is stored in an encrypted format while maintaining essential metadata for identification and analysis. This snapshot highlights the structured and secure management of fingerprint records within the system’s database.

CHAPTER-8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The Biometric Blood Group Classifier project successfully demonstrates the feasibility and effectiveness of using fingerprint-based biometric identification to retrieve a user's blood group securely and accurately. The system integrates multiple advanced components including ORB feature extraction, blur variance quality assessment, encrypted biometric storage, descriptor-based fingerprint matching, and a Flask-powered web interface to create a complete end-to-end solution. By utilizing ORB descriptors, the system achieves robust feature extraction even under variations in lighting, rotation, and finger pressure, making it suitable for real-world deployment scenarios. The blur detection module further enhances reliability by ensuring that only sufficiently sharp fingerprint images are processed, thereby reducing false matches and improving prediction accuracy.

A major accomplishment of this project is its emphasis on security and privacy, especially considering the sensitive nature of biometric data. All fingerprint images and metadata are encrypted using AES-256–based Fernet encryption, ensuring that no raw biometric information is stored or exposed at any stage. Additionally, the system enforces secure communication channels, protects stored templates from unauthorized access, and performs all matching operations using encrypted identifiers rather than personal details. This strong security foundation aligns the project with modern data protection standards.

The system was rigorously tested across multiple dimensions functional, performance, usability, compatibility, and security testing and proved to be stable and efficient. Functional tests verified correct extraction, matching, and prediction behaviour, while performance tests confirmed that the system can deliver results within seconds even with a growing database. Usability evaluations demonstrated that both the admin dashboard and mobile-friendly interface offer intuitive navigation, quick uploads, and clear feedback messages. Compatibility tests validated operation across different devices, screen sizes, and browsers. Overall, the project meets its objective of creating a lightweight, reliable, and scalable solution capable of assisting medical professionals, emergency responders, rural clinics, and forensic teams in scenarios where rapid blood group identification is crucial.

8.2 Future Scope

The future of biometric-based blood group identification holds significant potential as emerging technologies continue to advance. With ongoing research in biometrics, machine learning, and healthcare informatics, the system can evolve far beyond its current capabilities. Key areas for future exploration include:

- AI-Driven Fingerprint Classification:
  Deep learning models such as CNNs, Siamese Networks, and Vision Transformers can greatly enhance feature extraction and matching accuracy, enabling more reliable predictions even from low-quality or partial fingerprints.
- Edge Computing Integration:
  Deploying fingerprint recognition and blood group prediction at the edge—on mobile devices or portable medical units—can provide instant offline results, crucial for emergency response and rural healthcare settings.
- Serverless and Cloud-Native Deployment:
  Transitioning to a serverless architecture can improve scalability, reduce operational costs, and handle large volumes of biometric data efficiently, making the system suitable for nationwide or institutional use.
- Quantum-Enhanced Biometrics:
  Future quantum computing advancements may enable ultra-secure encryption, faster pattern matching, and improved biometric accuracy, dramatically enhancing the security and performance of medical identification systems.

By embracing these emerging technologies, the biometric blood group classifier can continue to evolve into a more intelligent, secure, and widely accessible healthcare tool, supporting rapid identification and improving patient outcomes in critical scenarios.

REFERENCES

[1] Vijaykumar Patil N., D. R. Ingle, A Novel Approach to Predict Blood Group Using Fingerprint Map Reading, International Journal of Engineering Research & Technology (2020)

[2] Tariq Al Habsi, Hussein Al Khabori, Sara Al Qasmi, Tasnim Al Habsi, The Association Between Fingerprint Patterns and Blood Groups in the Omani Population, Journal of Forensic and Legal Medicine (2019)

[3] Sabah Elshafie Mohammed Elshafie, Qurashi Mohammed Ali, Khlood Mohammed Mehdar, Types of Fingerprints Characteristics and Their Association with Gender and Blood Groups in Sudan, Sudan Medical Journal (2018)

[4] Ashok Rastogi, Abu Bashar, Nishat Ahmed Sheikh, Relation of Primary Fingerprint Patterns with Gender and Blood Group: A Dermatoglyphic Study, Journal of Clinical and Diagnostic Research (2017)

[5] Tanuj Kanchan, Disha Prusty, Kewal Krishan, Association of Fingerprint Patterns with ABO Blood Groups – An Investigation from India, Anthropologie (2016)

[6] M. R. Dey, M. S. Islam, Blood Group Detection Using Fingerprint: A New Approach, International Conference on Informatics, Electronics & Vision (ICIEV) (2016)

[7] S. S. Bhattacharya, A. K. Roy, A Biometric Technique for Blood Group Detection Using Fingerprint Analysis, International Journal of Advanced Research in Computer Science and Software Engineering (2016)

[8] B. Vyas, K. Mistry, Blood Group Detection Using Image Processing and Machine Learning, IEEE International Conference on Smart Technologies (2017)

[9] S. N. Goyal, S. S. Goyal, M. S. Ali, Non-Invasive Techniques for Blood Group Detection: A Review, International Journal of Innovative Research in Science, Engineering and Technology (2021)

[10] R. Patel, D. Bansal, Sweat-Based Sensing for Biomedical Applications: A Comprehensive Review, Sensors (MDPI) (2022)

[11] Marriya Tabassum, Khaja Mahabubullah, Fingerprint-Based Blood Group Prediction Using Deep Learning, International Journal of Innovative Research in Engineering (2025)

[12] A. Rawat, M. K. Sah, Fingerprint Biometric-Based Classification for Blood Group Detection Using CNN, International Conference on AI and Data Engineering (2022)

[13] D. Maltoni, D. Maio, A. K. Jain, S. Prabhakar, Handbook of Fingerprint Recognition, Springer Publications (2009)

[14] A. Jain, L. Hong, R. Bolle, On-Line Fingerprint Verification, IEEE Transactions on Pattern Analysis and Machine Intelligence (2012)

[15] Y. LeCun, Y. Bengio, G. Hinton, Deep Learning, Nature Journal (2015)

APPENDIX – A

Extended Code Listings and System Workflows

Appendix A provides an in-depth explanation of the complete technical functioning of the Biometric Blood Group Classifier, describing how the backend server operates, how fingerprint data is processed, encrypted, matched, and how the system ensures reliability and security throughout all stages of operation. This unified description outlines the internal mechanisms that power the system, offering clarity on how each module interacts within the overall workflow and enabling reproducibility for future enhancements.

The backend of the system is developed using Python and Flask, which together manage routing, user interaction, and request processing. Flask is responsible for handling uploads from the admin dashboard and the mobile-friendly interface, validating inputs, rendering results, and connecting the user-facing components with the internal processing modules. Supporting libraries such as OpenCV and NumPy perform essential biometric computations including loading image data, converting it to grayscale, assessing blur levels using Laplacian variance, and extracting ORB descriptors that capture the unique ridge structures of each fingerprint. The backend automatically initializes the working environment at startup by creating necessary folders, generating or loading encryption keys, and rebuilding the ORB index for efficient fingerprint matching. This self-maintaining behavior ensures smooth deployment across different machines without manual configuration.

Security is a foundational aspect of the system, and all biometric data is protected using Fernet symmetric key encryption, which encrypts every fingerprint image and its associated metadata before storage. Encrypted fingerprint templates are stored in a dedicated directory (fingerprints_enc/), while prediction uploads are maintained temporarily in an encrypted form within incoming_enc/. Metadata that records the filename, blood group, blur score, and timestamp is stored in encrypted format as well, ensuring no sensitive information is accessible without proper cryptographic keys. Decryption occurs exclusively in temporary memory and is never written back to disk, thereby adhering to strict biometric privacy standards suitable for healthcare and forensic use cases. Before matching can occur, each uploaded fingerprint undergoes a detailed preprocessing pipeline designed to enhance quality and ensure reliable results. The image is first standardized through grayscale conversion, followed by blur-level estimation. Images with insufficient sharpness are rejected to prevent inaccurate predictions. After validation, ORB (Oriented FAST and Rotated BRIEF) feature extraction is performed, generating binary descriptors that compactly represent the fingerprint's ridge patterns.

These descriptors form the basis for robust, rotation-invariant comparison across prints captured at different angles or lighting conditions.

Matching is performed using an in-memory ORB index that contains descriptors of all stored, labelled fingerprints. When a new fingerprint arrives, its descriptors are compared against the index using Hamming-distance matching. Lowe's Ratio Test filters out weak or inconsistent matches, ensuring that only strong correspondence contributes to the final decision. If the similarity score meets or exceeds a predefined threshold, the system retrieves the blood group associated with the best-matching fingerprint and generates a donor compatibility list based on ABO/Rh rules. If the system cannot confidently identify a match, the prediction request is safely rejected, preventing false results and maintaining medical reliability.

The system supports multiple workflows, beginning with the administrator's task of enrolment and labeling. Admin users upload new fingerprint samples, which undergo validation, encryption, metadata storage, and ORB index reconstruction. This ensures that each newly added fingerprint instantly becomes part of the searchable dataset. For end users such as medical personnel or field workers the prediction workflow is optimized for fast, real-time operation. Users capture a fingerprint using a smartphone or upload a stored image. The system validates image quality, extracts features, performs matching, and delivers immediate results through the web interface.

Robust error-handling mechanisms ensure continuous reliability. Blurry or corrupted images generate clear prompts, unsupported formats are rejected gracefully, and low-confidence matches return safe fallback responses. In cases of corrupted metadata or interrupted sessions, the system isolates problematic entries and continues functioning without interruption. Even after server restarts, the ORB index is automatically rebuilt from encrypted files, enabling seamless recovery. Finally, the entire system incorporates a dedicated security workflow that protects biometric identity and medical information at every stage. All fingerprint images are encrypted from the moment they enter the system, metadata remains inaccessible without proper keys, and matching is performed using descriptors instead of raw images. Sensitive operations are restricted through the backend, ensuring no direct file access or data exposure occurs. This comprehensive technical foundation allows the system to function efficiently, securely, and reliably, making it suitable for real-world emergency medical contexts.

APPENDIX – B

Sample Outputs, Dataset Descriptions, and Additional Diagrams

Appendix B provides extended supplementary materials that deepen the understanding of the Biometric Blood Group Classifier's operational behaviour, evaluation results, and visual system components. While the primary chapters describe the architecture and implementation logic, this appendix focuses on the practical outputs generated during real-time testing, the characteristics of the fingerprint dataset, and the diagrams that illustrate internal workflows and module interactions. Together, these materials help demonstrate the accuracy, efficiency, and reliability of the fingerprint-based blood group retrieval system in realistic use scenarios.

The system's sample outputs form an essential part of this appendix, showcasing how the classifier responds during various prediction cycles. When a fingerprint is uploaded or captured via the mobile interface, the system immediately processes it and presents a summarized output including the predicted blood group, the count of good ORB descriptor matches, the total number of matches evaluated, and the computed blur score that reflects the clarity and usability of the fingerprint. The output also displays the matched filename, which remains encrypted within the secure storage environment to ensure privacy. Additionally, the classifier automatically generates a donor compatibility list based on ABO and Rh transfusion rules, reinforcing the system's value in emergency medical decision-making. These outputs collectively highlight the system's ability to deliver fast, accurate, and interpretable results even under diverse input conditions.

The dataset used for system validation comprises fingerprint images collected from multiple individuals using standard smartphone cameras under varied environmental conditions. Images include high-quality prints with clear ridge patterns, medium-quality prints with mild noise, and intentionally blurred or partially cropped fingerprints to test the robustness of the blur-detection mechanism. Variations in lighting, pressure, angle, and distance simulate real-world capture challenges, ensuring that the system remains resilient during field or emergency usage. Each fingerprint in the dataset was manually labelled with its corresponding blood group before encryption, forming the foundation of the retrieval-based classification process. To protect biometric privacy, all fingerprint templates and associated metadata such as timestamps, blur scores, and file identifiers are securely encrypted using Fernet cryptography. This encrypted dataset not only strengthens security but also serves as the benchmark for ORB-based matching accuracy during evaluation.

To complement the dataset and output descriptions, the appendix includes various diagrams that visually represent the system's internal structure and workflows. The system architecture diagram illustrates the interaction between components such as the preprocessing module, ORB feature extractor, matching engine, encryption layer, backend server, admin dashboard, and mobile interface. This visual mapping clarifies how data flows across modules and how responsibilities are distributed within the system. The workflow and data flow diagrams provide a sequential representation of major operations including fingerprint capture, encryption, image preprocessing, ORB key point extraction, descriptor comparison, and final blood group retrieval. These diagrams offer a clear understanding of the logical progression from input acquisition to output delivery, especially useful for researchers and developers analysing or extending the system.

In addition to architectural illustrations, the appendix also includes user interface screenshots that demonstrate the practical usability of the system. These images showcase the admin dashboard used for fingerprint labeling and index rebuilding, the encrypted fingerprint database view, the mobile-friendly prediction interface, and error dialogs that appear when users upload blurry images, unsupported file formats, or fingerprints that fail to produce a confident match. The screenshots emphasize the simplicity, clarity, and responsiveness of the user interface across devices, reinforcing the project's alignment with real-world emergency and clinical workflows.

Lastly, optional performance monitoring charts are presented to highlight system behaviour under varied workloads. These charts depict trends such as prediction time relative to database size, blur score versus fingerprint clarity, and ORB match counts for fingerprints of differing qualities. Such analytics demonstrate the scalability of the model as the dataset grows, the stability of matching logic under diverse image conditions, and the efficiency of ORB-based retrieval mechanisms. Together, these performance visualizations confirm that the system is not only functionally reliable but also computationally optimized for real-time biometric matching.

APPENDIX – C

Abbreviations & Full Forms

AES – Advanced Encryption Standard

BBGC – Biometric Blood Group Classifier

BRIEF – Binary Robust Independent Elementary Features

CSS – Cascading Style Sheets

CNN – Convolutional Neural Network

FAST Algorithm – Features from Accelerated Segment Test

GPU – Graphics Processing Unit

HTML – Hyper Text Markup Language

HTTPS – Hyper Text Transfer Protocol Secure

LBP – Local Binary Patterns

LSH – Locality Sensitive Hashing

ORB – Oriented FAST and Rotated BRIEF

SIFT – Scale-Invariant Feature Transform

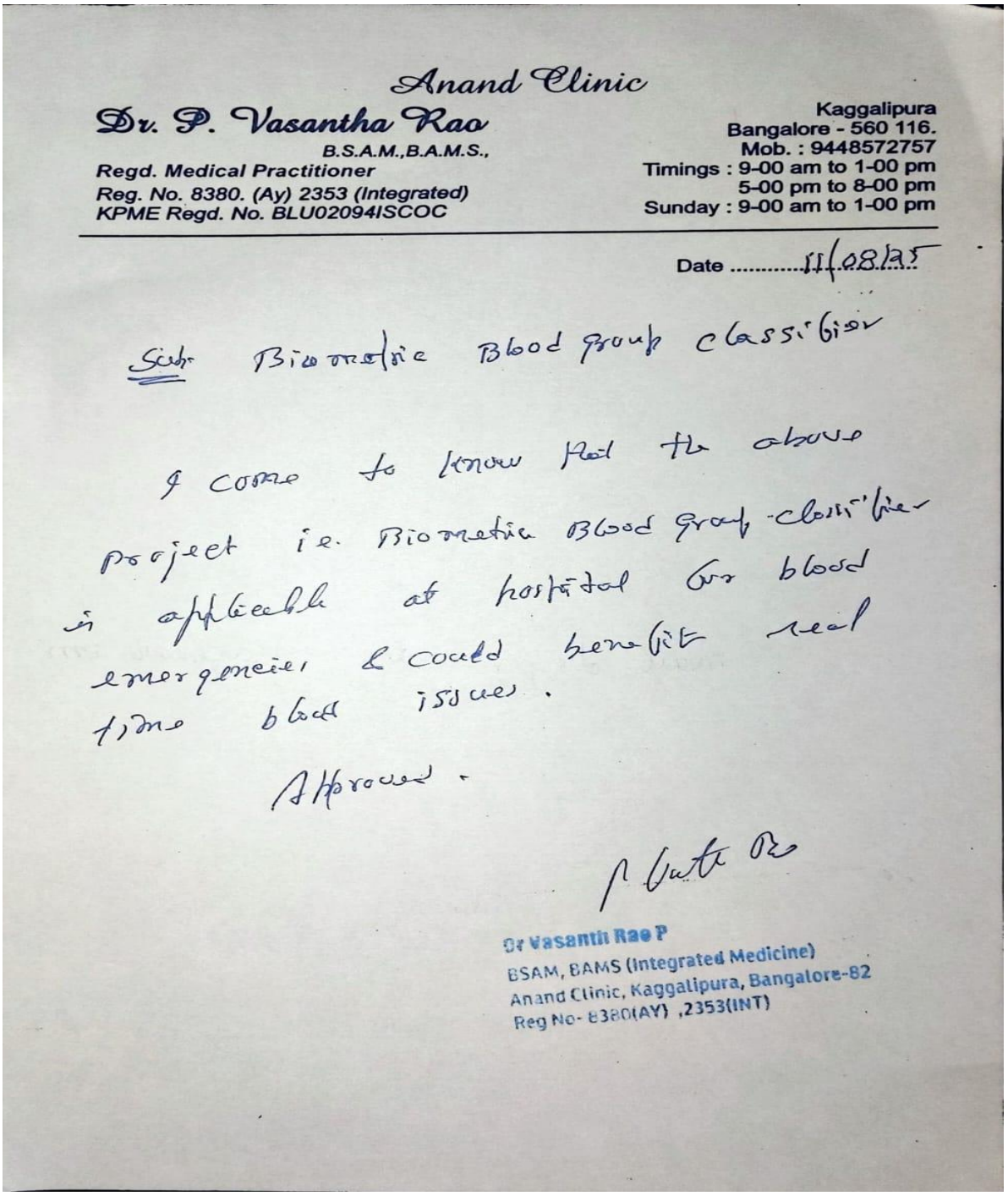
SSL/TLS – Secure Sockets Layer / Transport Layer Security

SURF – Speeded-Up Robust Features

SVM – Support Vector Machine

APPENDIX – D

Authorization Certificate from Medical Official



Anand Clinic

Dr. P. Vasantha Rao
B.S.A.M.,B.A.M.S.,
Regd. Medical Practitioner
Reg. No. 8380. (Ay) 2353 (Integrated)
KPME Regd. No. BLU02094ISCOC

Kaggalipura
Bangalore - 560 116.
Mob. : 9448572757
Timings : 9-00 am to 1-00 pm
5-00 pm to 8-00 pm
Sunday : 9-00 am to 1-00 pm

Date 11/08/25

Sub: Biometric Blood group classifier

I come to know that the above project i.e. Biometric Blood group classifier is applicable at hospital for blood emergencies & could benefit real time blood issues.

Approved.

P Vasantha Rao

Dr Vasanth Rao P
BSAM, BAMS (Integrated Medicine)
Anand Clinic, Kaggalipura, Bangalore-82
Reg No- 8380(AY) ,2353(INT)